# Deep Feature Enhancement Method for Land Cover With Irregular and Sparse Spatial Distribution Features: A Case Study on Open-Pit Mining

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***Abstract*— Land cover classification in mining areas (LCMA) is essential for the environmental assessment of mines and plays a crucial role in their sustainable development. The shapes of mine land occupation elements are irregular, and the overall proportion of their area is relatively small. Therefore, their features may be easily lost during feature extraction, which limits the interpretation accuracy in mining areas. This study attempts to address these issues. We propose a model named EG-UNet to enhance the features of elements with few samples and to capture long-range information. The proposed EG-UNet includes two main modules. First, the edge feature enhancement module, the edges of elements of mine land occupation contain more information than other spatial locations. Hence, during the feature extraction of elements, a Sobel operator is used to extract the object boundary, which increases the weight of these features before the pooling operation for their preservation. Second, the long-range information extraction module, long-range information helps extract tiny objects, such as dumping grounds in the mining area. We present a graph convolutional network (GCN) to capture the long-range features and apply convolutional neural networks to learn the graph construction. A total of ten deep-learning networks were compared using the LCMA semantic segmentation dataset. Our model exhibited the best performance, especially in classifying classes with few samples. Furthermore, to evaluate the general ability of EG-UNet, a benchmark-Gaofen Image Dataset (GID) was used, and the result still reflected the superiority of our method.**

***Index Terms*— Convolutional neural network, graph convolutional network (GCN), open-pit mining, remote sensing, semantic segmentation, U-Net.**

## I. Introduction

LAND cover has undergone unprecedented changes with economic and industrial development. Consequently, the irresponsible abuse of land is driving land degradation, which is regarded as one of the most destructive effects on the terrestrial environment [1], [2], [3]. Several studies have revealed that open-pit mining significantly influences local land degradation [1], [2], [3], [4], [5], [6], [7], [8]. Environmental problems, such as water pollution, air pollution, and the destruction of biodiversity, occur during the development and beneficiation of open-pit mining areas [9], [10], [11], [12], [13].

In recent years, the construction of mine areas has been emphasized to reduce the land degradation and biodiversity impact resulting from mining areas. Detailed land cover classification is essential for mining areas' sustainable development and construction [12], [14], [15], [16], [17].

Compared with other land cover types, the mining area is relatively remarkable. When we consider the mine surface elements as a single class, it is irregular (including shape and size), scattered, and covers a small area [see Fig. 1(a)], which is different from farmland, building, and other common land covers. Furthermore, for detailed semantic segmentation in the mining area, the previous characteristics are more evident in some mining area targets, such as the dark blue object in Fig. 1(b).

Land cover classification in mining areas (LCMA) is an essential method for assessing the mitigation measures that are implemented in the environment [13]. However, a lack of detailed semantic segmentation datasets and specially designed models in mining still challenges LCMA.

### A. Datasets for LCMA

Remote sensing data sources for LCMA have progressed from low resolution to high resolution. Landsat satellite data have been used since the 1970s, and various researchers have focused on the classification of tailings and mine areas using hyperspectral data [18], [19], [20], [21]. However, the limited data sources for open-pit areas and the narrow swath width restrict the application of hyperspectral imagery [13]. With the development of remote sensing technology, some researchers have tried to classify land cover in mining areas with medium- and high-resolution satellite imageries [22], [23], [24], [25], [26], [27], [28], [29]. Lobo et al. [25] mapped mining areas within 13 regions of the Brazilian Amazon with Sentinel-2 images. In addition, several studies have performed LCMA using Beijing-1, GF-2, and Landsat TM.

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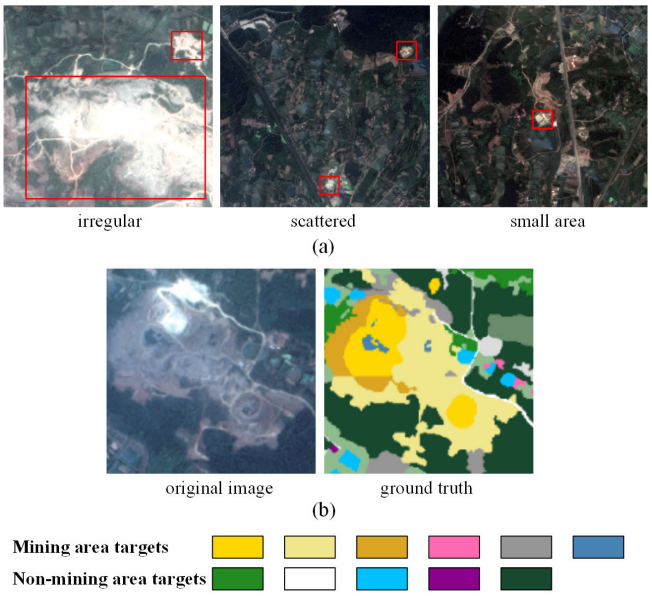


Fig. 1. Characteristics of land covers in the mining area. (a) Mining area and nonmining area (red boxes circle the mining area). (b) Detailed semantic segmentation in the mining area.

Although the previous studies constructed their datasets, they only contained some sample points in the study area. In addition, the spatial and context information is lost when classifying each sample point, which restricts the classification accuracy, especially for objects with sparse spatial distributions and irregular shapes in mining areas. Furthermore, the number of samples limits the application of deep learning [30]. To increase the amount of data, Xie et al. [28] constructed a semantic segmentation dataset of all pixels in an open-pit mining area; however, they considered the mine surface elements as a single class. In this case, less information on mining areas was provided. With the promotion of sustainable mine construction, such coarse classification schemes in mining areas cannot meet the requirements of ecological mine construction [13]. Therefore, it is necessary to construct a fine-scale land cover classification semantic segmentation dataset for mining areas.

### *B. Related Methods*

The aim of land cover classification with remote sensing imagery is to categorize each pixel into predefined land cover classes. Machine learning is essential for classifying different categories from remote sensing datasets. Various algorithms have been used extensively owing to their excellent performance, such as the support vector machine (SVM) [2], [31], the artificial neural network (ANN) [32], random forest (RF) [33], and k-nearest neighbors [34]. Several studies have demonstrated that the SVM exhibits the best accuracy in classifying surface-mined land, followed by the RF and ANN; in contrast, RF is the best in mapping surface-mined and agricultural landscapes, followed by the SVM and ANN [2], [35], [36], [37], [38], [39]. However, the accuracy of machine learning is strongly dependent on the set of hyperparameters [40], [41], [42], and the feature selection methods that are applied to select and embed features in the classification process also affect the accuracy of the classification algorithm [38], [43], [44]. Moreover, owing to the complexity of mining areas, it is difficult to apply a classification method to depict the features of mining areas in remote sensing imagery [28].

In recent years, deep convolutional neural networks (DCNNs) have attracted substantial attention in the area of land cover classification with high-resolution remote sensing imagery [12], [45], [46], [47], [48] owing to their strong capability for extracting high-dimensional and semantic information from images [49], [50]. CNNs are end-to-end models that can automatically label the land cover classification in remote sensing imagery without additional classifiers or postprocessing. Although some CNN structures, such as the fully convolutional network (FCN) [51], U-Net [52], and SegNet [53], have achieved satisfactory results in terms of accuracy for land cover classification, the lack of long-range information available in the images limits the performance of the models. To address this problem, DeepLab [54] applies dilated convolutions to expand the receptive field of CNNs. However, dilated convolution is dominated by large objects and exhibits poor performance for small objects [55]. Graph convolutional networks (GCNs) [56] and transformers [57] performed well in long-range information capturing. The gated graph CNN classifier [58], which uses GCNs [56], has achieved state-of-the-art accuracy in building segmentation. Swin transformer [59] designed a hierarchical transformer with shifted windows, and its performance surpasses the previous state-of-the-art on COCO [60]. Chen et al. [61] proposed a hierarchical transformer with an edge-aware module to capture long-range context information for building extraction. Yuan et al. [62] used a binary relation matrix to represent the relationship between all pixels, similar to GCN, and applied self-attention to capture contextual information. Mou et al. [63] proposed spatial and channel relation modules for contextual information modeling and achieved competitive results. In recent studies of land cover classification, Li et al. [64] fused the features extracted by different convolutional networks to enhance the expression ability of the features, and the proposed model achieved competitive results compared to state-of-the-art models. Fan et al. [65] adopted an automatic sample selection method to enhance the dataset, and experiments confirmed the superior performance of the proposed model. Xie et al. [28] combined the characteristics of UNet, SegNet, and D-LinkNet [66] to propose DUSegNet, which outperformed other common deep learning models. Zhang et al. [67] adopted and modified a high-resolution network (HRNet) [68] to produce high-resolution features without the decoding stage and to preserve additional spatial details. Zhang et al. [69] combined the strengths of residual learning and U-Net to develop a model known as ResUNet. They demonstrated that the rich skip connections within ResUNet could facilitate information propagation.

Although these studies presented development strategies, in the mining area, the shapes of objects, such as opencast pits and dumping grounds, are irregular and sparsely distributed,

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which makes it difficult to propagate features of the same type of objects. It will result in a weak representation of extracted features. Furthermore, compared with common land cover objects (e.g., forestland and farmland), the number of pixels of an opencast pit is fewer. Hence, when applying DCNNs, especially in pooling operations, their features are more likely to be lost, resulting in low classification accuracy [70], [71], [72]. Pooling operation aims at feature compression. Therefore, it is difficult to ensure that the critical information of mining area objects is preserved during the dimension reduction process.

### C. Contributions

To solve these problems, this study constructed a fine-scale land cover semantic segmentation dataset of open-pit mine areas with ZY-3 satellite imagery, which contained seven fine-scale open-pit mining area objects and 13 common land cover objects, to compensate for the current lack of a fine-scale LCMA dataset. In addition, we proposed a deep learning model named EG-UNet, for LCMA, especially for objects with irregular shapes and sparse spatial distributions in the mining area.

The innovations of the research are given as follows.

1) To avoid the problem of feature loss in extracting features of mining areas, we constructed an edge feature enhancement (EE) module to enhance the boundary features before the pooling layer. The boundaries of elements of mine land occupation contain more information and affect the classification accuracy [49]; therefore, in this manner, the edge features can be better preserved following enhancement, especially for classes with few samples.
2) To solve the problem of weak representation of extracted features, we used a GCN module to capture long-range information between objects of the same type. A CNN model was used to learn the graph construction of features.

To further evaluate the general capabilities of the proposed model, we applied Gaofen Image Dataset (GID) [73] as a benchmark. Nine deep learning methods were used for a performance comparison with EG-UNet, including the classical models: U-Net, UNet++, Attention U-Net [74], DeepLab v3+, PSPNet [75], HRNet, and ResUNet, and the latest models: SwinUNet [76] and DE-DCGCN-EE [77]. Between these latest models, SwinUNet applied Swin Transformer as the encoder to extract context features for medical image segmentation. DE-DCGCN-EE, similar to the proposed models, designed an edge enhancement module to preserve the edge of blood vessels and applied GCNs to utilize channel information sufficiently.

## II. Methods

The proposed model, EG-UNet, is based on U-Net and includes several improvements. U-Net is a classic semantic segmentation model that can be divided into the encoder and decoder modules. The input image is downsampled four times in the encoder part by a max pooling operation to capture the high-level features. In the decoder part, these features are upsampled four times by the transposed convolution operation; before upsampling, the corresponding output in the encoder, which has the same size as the current features, is fused.

EG-UNet retains the main structure of U-Net but includes two modules to improve the extraction accuracy (see Fig. 2) as follows.

1) *EE Module:* This module enhances the boundary features and is used before the max pooling layer to reduce the impact of the pooling process on the boundary information.
2) *Long-Range Information Module:* GCNs are applied to capture the global contextual information, and this module is used after the final downsampling operation owing to the available computing power.

### A. EE Module

The boundaries of objects contain more information than other spatial locations, which is essential for classifying each pixel [46]. However, these important features may be lost following max pooling. Hence, we use the Sobel operator [78] to extract the position of the objects' edges and, subsequently, strengthen these features located on the edge, to improve their preservation rate.

Fig. 3 depicts the structure of the EE module. There are two categories of objects. $X_{\text{in}}$ represents the input features, with a dimension of $D$. The light blue pixels belong to object a, whereas the pink pixels belong to object b. First, the Sobel operator, which consists of $G_x$ and $G_y$, is used to detect the edges of these objects in each dimension. $G_x$ is a $3 \times 3$ filter and convolutes in the $X$-direction, while $G_y$ convolutes in the $Y$-direction; therefore, the edge matrices containing the edge locations are obtained as follows:

$$M^i = \left(G_x\left(X_{\text{in}}^i\right) + G_y\left(X_{\text{in}}^i\right)\right), \quad i \in [1, D]. \tag{1}$$

Then, morphological closing operations $C$, which consists of dilation and erosion operations, are used to remove the noise, connect broken edges in $M$, and obtain new edge matrices $M'$ as follows:

$$M'^i = C\left(M^i\right), \quad i \in [1, D]. \tag{2}$$

Subsequently, the input features are used to multiply $M'$, and the result ($X_{\text{out}}$) represents the output features of the EE module. The red pixels represent the enhanced pink pixels (object b), whereas the dark blue pixels represent the enhanced blue pixels (object a)

$$X_{\text{out}} = M'^i * X_{\text{in}}^i, \quad i \in [1, D]. \tag{3}$$

In (5), the "$*$" operation denotes multiplying the values of the corresponding positions between $M'^i$ and $X_{\text{in}}^i$. As shown in Fig. 3, object b has fewer pixels than object a in $X_{\text{in}}$, but, in $X_{\text{out}}$, the enhanced pixels mainly come from object b. Hence, the EE module is effective for classes with fewer samples. After max pooling, these enhanced features are better preserved.

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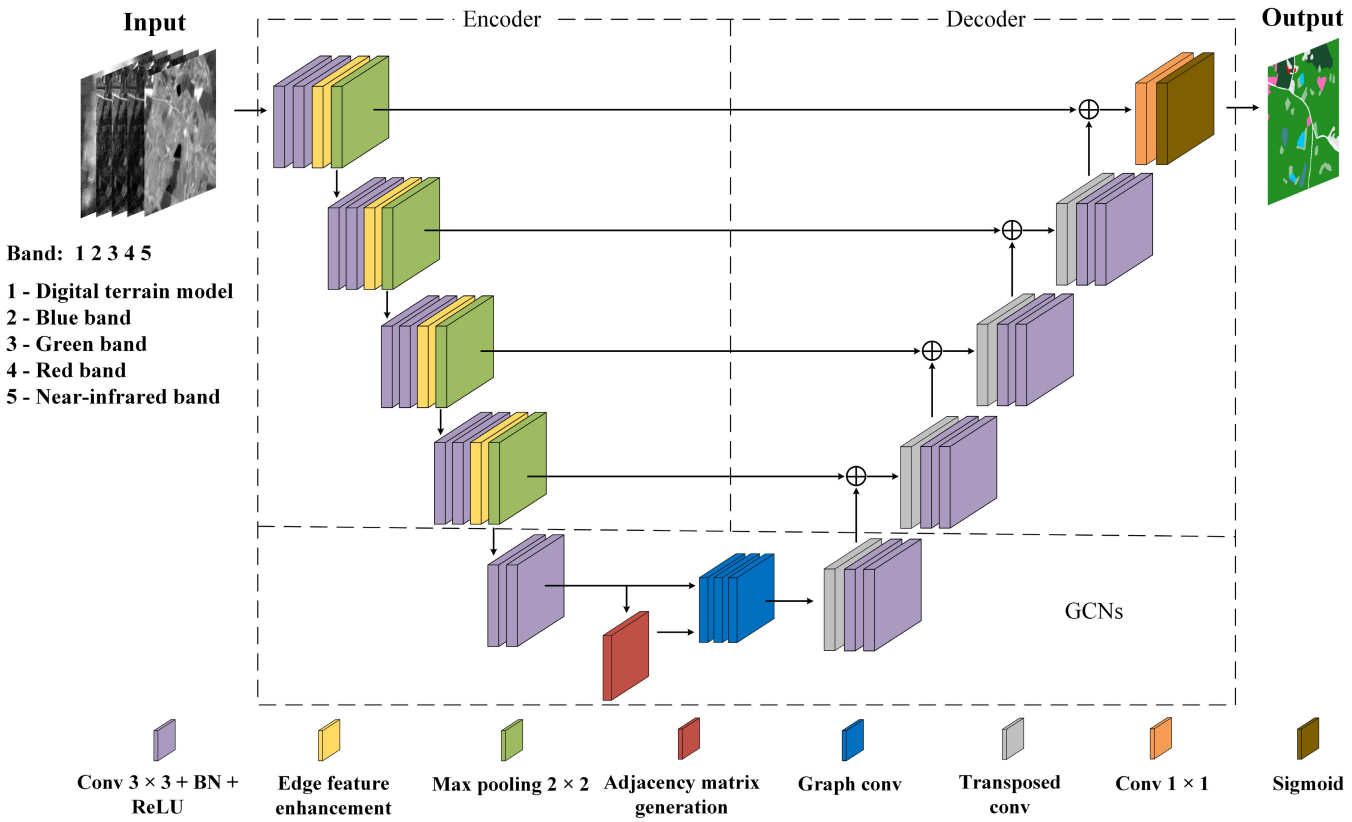


Fig. 2. Framework of EG-UNet.

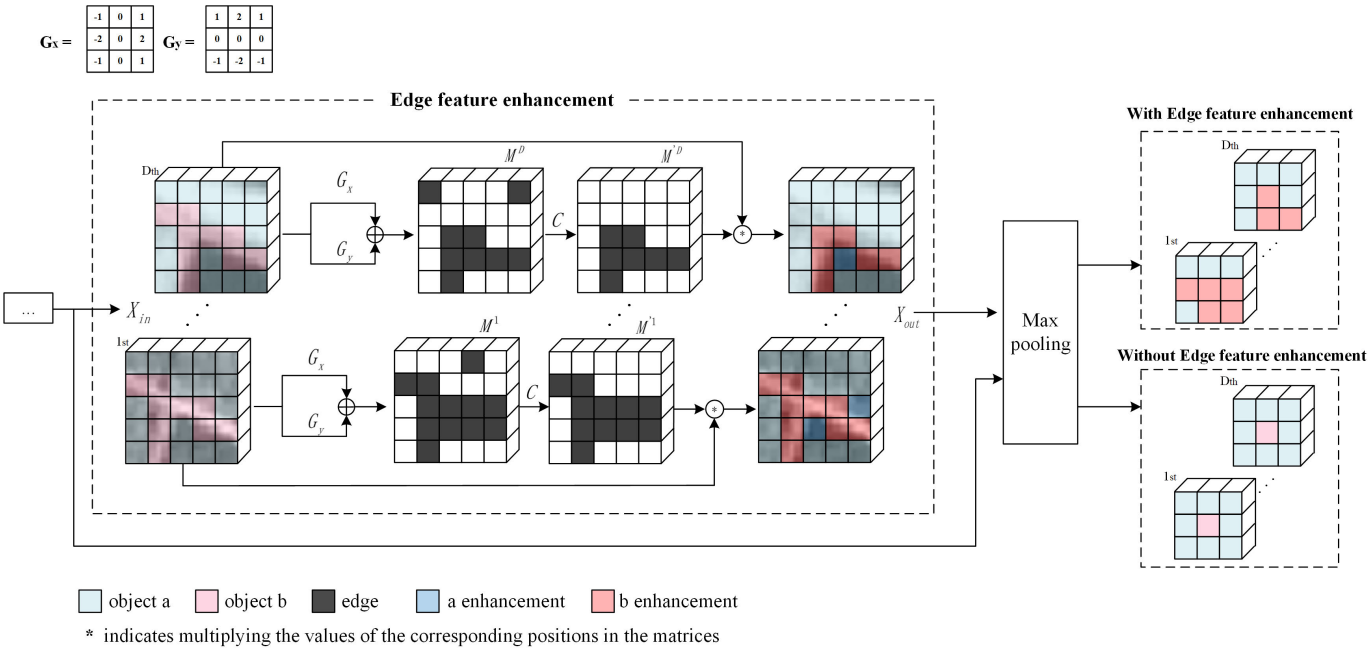


Fig. 3. Structure of the EE module.

### *B. Long-Range Information Module*

Assume an input tensor $X \in \mathbb{R}^{N \times D}$, where $D$ is the feature dimension and $N$, which is equal to $H \times W$, is the number of locations defined on the grid coordinates. In regular convolution, information is only exchanged in a small neighborhood, depending on the filter size (typically, $3 \times 3$). GCNs extend the concept of convolution to graph-structured data through the construction of an adjacency matrix. Although GCNs were initially applied in knowledge graphs, in recent years, they have been used in the feature extraction of natural images. GCNs can enlarge the receptive field and capture contextual information compared to the limited receptive fields of other

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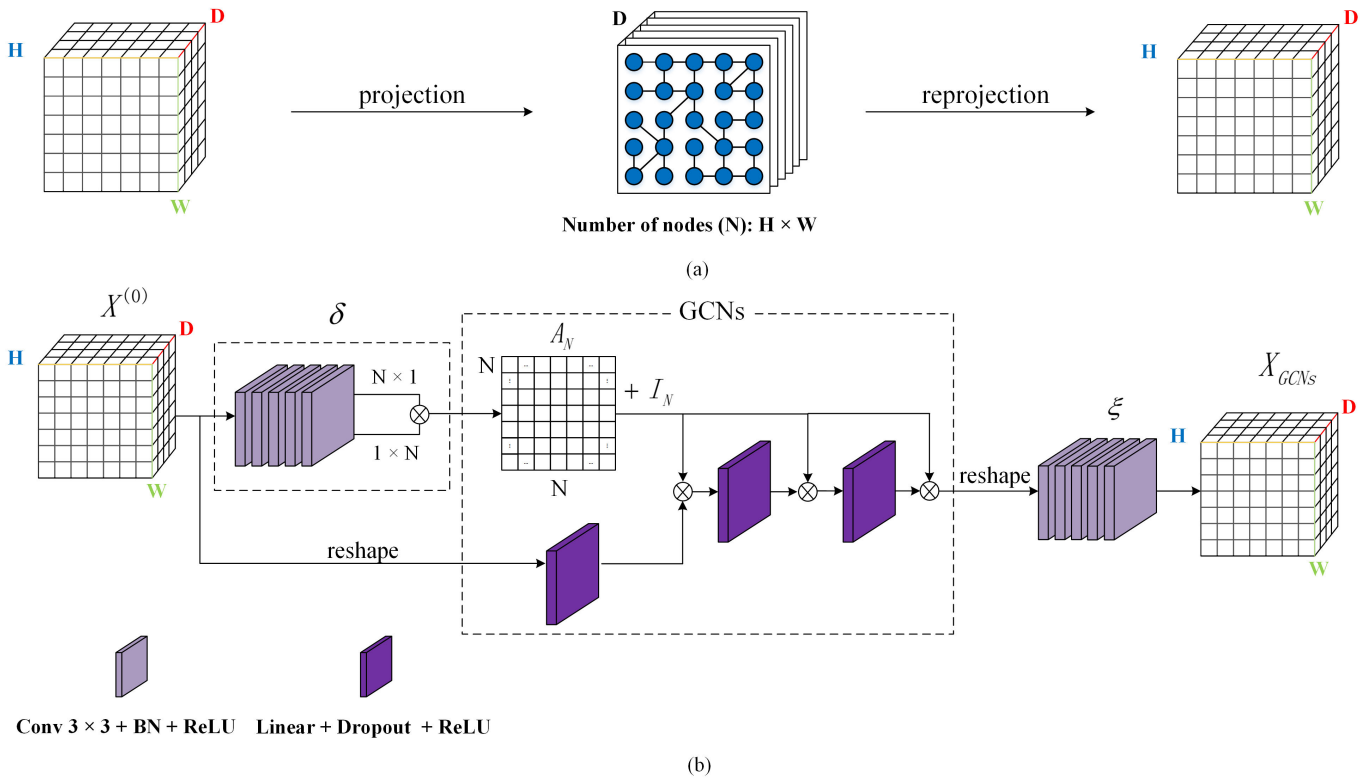


Fig. 4. Structure of the long-range information module: (a) sketch and (b) details.

CNNs [58], [72], [80]. Graph convolution is effective and efficient because, unlike standard convolution, it generalizes the neighborhood definition and allows for long-range information exchange in a single layer. Following [56], by defining an undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with $N$ nodes $v_i \ \epsilon \mathcal{V}$ and edges $(v_i, v_j)\epsilon\mathcal{E}$, the graph convolution can be defined as

$$X^{(l+1)} = \sigma\left(\tilde{D}^{-\frac{1}{2}}\tilde{A}\tilde{D}^{-\frac{1}{2}}X^{(l)}W\right) \tag{4}$$

where $\sigma$ denotes a nonlinear activation function, $\tilde{A} = A_N + I_N, \epsilon \mathbb{R}^{N\times N}$ is the adjacency matrix of undirected graph $\mathcal{G}$ with added self-connections, $\tilde{D}$ is the degree matrix of $\mathcal{G}$, $\tilde{D}_{ii} = \sum_j A_{ij}$, $W$ is the layer-specific trainable weight matrix, and $I_N$ is the identity matrix.

The definition of the graph structure is vital for determining information propagation. The proposed framework projects the input features $(X^{(0)})$ to a new dimension by operation $\delta$ and learns the construction rules of the adjacency matrix $A_N$ [see (5)]. Then, apply operation $\xi$ to reproject these graph features into the original size [see (9)]

$$A_N = \delta(X^{(0)}) \tag{5}$$

$$X_{\text{GCNs}} = \xi(X^{(l+1)}). \tag{6}$$

In (5), operation $\delta$ changes the size of $X^{(0)}$ and generates the adjacency matrix $A_N$. In (9), operation $\xi$ resizes $X^{(l+1)}$ to the size of $X^{(0)}$. $\delta$ and $\xi$ both consist of five convolutional layers (the filter size is 3 × 3). In operation $\delta$, after the final convolutional layer, the feature size is reshaped to $N \times 1$ and $1 \times N$, respectively. Subsequently, $A_N$ is obtained by multiplying these two matrices. Fig. 4 depicts the structure of the GCNs.

The proposed EG-UNet model uses three graph convolution layers to calculate the feature representation. According to [56], three layers provide the best results when GCNs are used to capture long-contextual information.

## III. Fine-Scale LCMA Dataset

### A. Study Area

The study area is located in Jiangxia district, Wuhan city, Hubei province, China, and covers an area of 109.4 km$^2$ (114°12′ 33.59″ E–114°23′ 6.89″ E and 30°15′ 38.85″ N–30°18′ 57.48″ N). Numerous mineral resource-relevant activities take place in the study area. The mine disturbance owing to open-pit mining has a history of nearly 70 years, and most mines are currently active. Fig. 5 depicts the study area.

A ZiYuan-3 (ZY-3) stereo satellite image acquired on June 20, 2012, was used in the analysis. ZY-3 has four cameras: one 2.1-m nadir-looking panchromatic camera, two 3.6-m front- and backward-looking panchromatic cameras, and one 5.8-m nadir-looking multispectral camera. The 3.6-m resolution front- and backward-looking panchromatic data were used to extract the relative digital terrain model (DTM) data with a resolution of 10 m using the Environment for Visualizing Images 5.0 software. After that, 2.1-m resolution panchromatic–multispectral fused data were generated (16 bit) [35], [72], which were used as the input features for the proposed model.

A fine-scale remote sensing classification system for surface cover information in mining areas was established regarding

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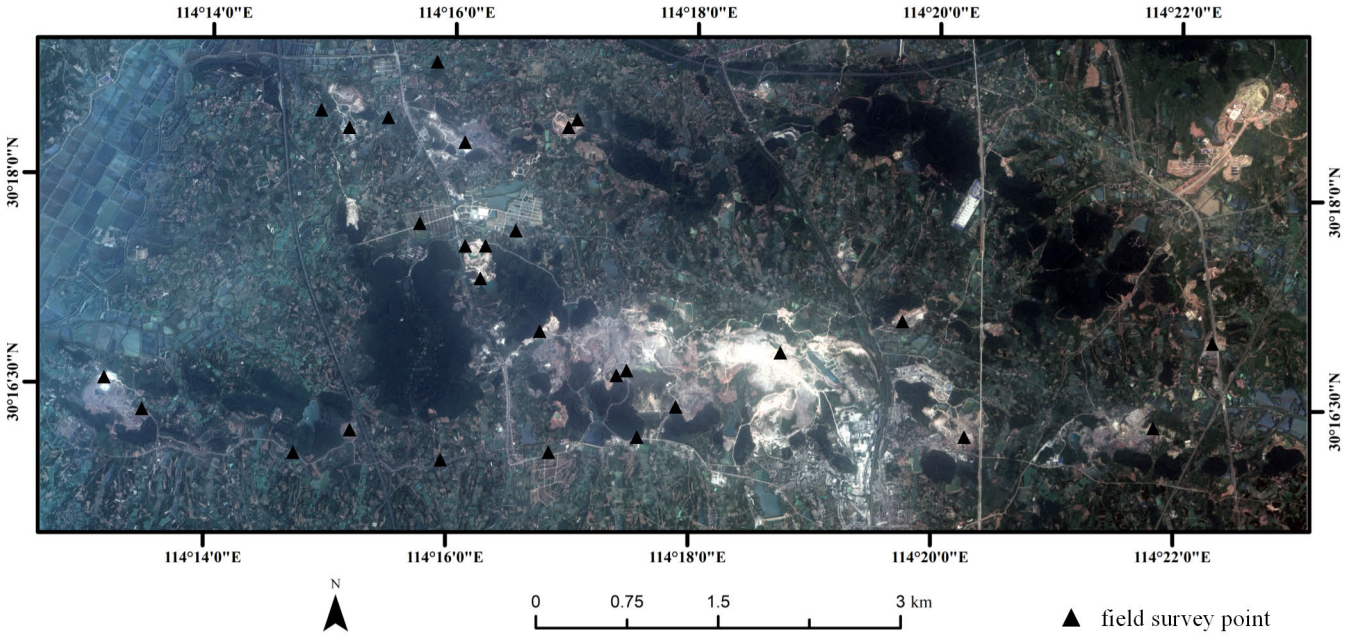


Fig. 5. Location of the study area and field survey samples, with ZY-3 fused true color images (R: red, G: green, B: blue) [38].

TABLE I
LAND COVER CLASSIFICATION SCHEMES USED IN THIS STUDY (REVISED FROM [2])

| No. | Fine land cover type | Description |
|---|---|---|
| 1 | Opencast pit | The mining operation area, including mined and abandoned mining pits |
| 2 | Ore processing site | The beneficiation operation activity area, including ore stacking and beneficiation product stacking areas |
| 3 | Dumping ground | A place where the stripped soil and covering materials on the surface layer are stacked together in the mine exploration and development process |
| 4 | Paddy field | An area with an adequate water supply that is used for the cultivation of rice, lotus, and other aquatic crops |
| 5 | Vegetable and fruit greenhouse | A structure with white plastic film sides and roofs and high surface albedo with regular rectangular shapes |
| 6 | Green dry land | Cultivated land without irrigation water sources and facilities that rely on natural precipitation for crops to grow |
| 7 | Gray dry land | A portion of arable land that is left idle for sowing |
| 8 | Nursery and orchard | An area with a rectangular shape like cropland that is dotted with vegetation cover and exposed soil and is black in true color images |
| 9 | Woodland | This includes timber stands, economic forests, and shelterbelts that have a high chlorophyll content and is dark red in false-color images |
| 10 | Forest under stress | An area that is under the influence of surface mining development is located around surface-mined land, has large amounts of deposited mineral dust, exhibits poor growth, and is grayish in true-color images |
| 11 | Shrub forest | Such a forest contains multiple stems with shorter heights, generally less than 2 m tall, and it is bright red in false color images |
| 12 | Pond and stream | This includes many fish ponds with regular rectangular shapes |
| 13 | Mine pit pond | A lake that is created during and after mining, typically with an irregular shape |
| 14 | Dark road | A black road in true color images, which usually refers to an asphalt highway |
| 15 | Bright road | A white road in true color images, which usually refers to a cement road |
| 16 | Light gray road | A gray road in true color images, which usually refers to a dirt road |
| 17 | Red roof | Usually refers to rural land |
| 18 | Bright roof | Usually refers to urban and town areas |
| 19 | Blue roof | Usually refers to land that is used for industrial parks |
| 20 | Bare surface | Exposed land with little vegetation |

the land cover classification system proposed by the Food and Agriculture Organization of the United Nations and the land cover classification system of the resources and environment database of the Chinese Academy of Sciences. Based on previous studies in this area [2], [10], [38], seven fine-scale open-pit mining area classes and 13 common land cover classes were used in this work (see Table I for details).

We manually interpreted the study area through field investigations and previous research [2], [10], [38]. Fig. 6 presents the results of the visual interpretation. In addition, we set up 29 field survey points to verify the interpretation results (see Fig. 4).

### B. LCMA Semantic Segmentation Dataset

The study area was divided into 32 areas in longitude and 13 areas (from A to M) in latitude according to a patch size of 256 × 256. An 87.1% area duplication existed between

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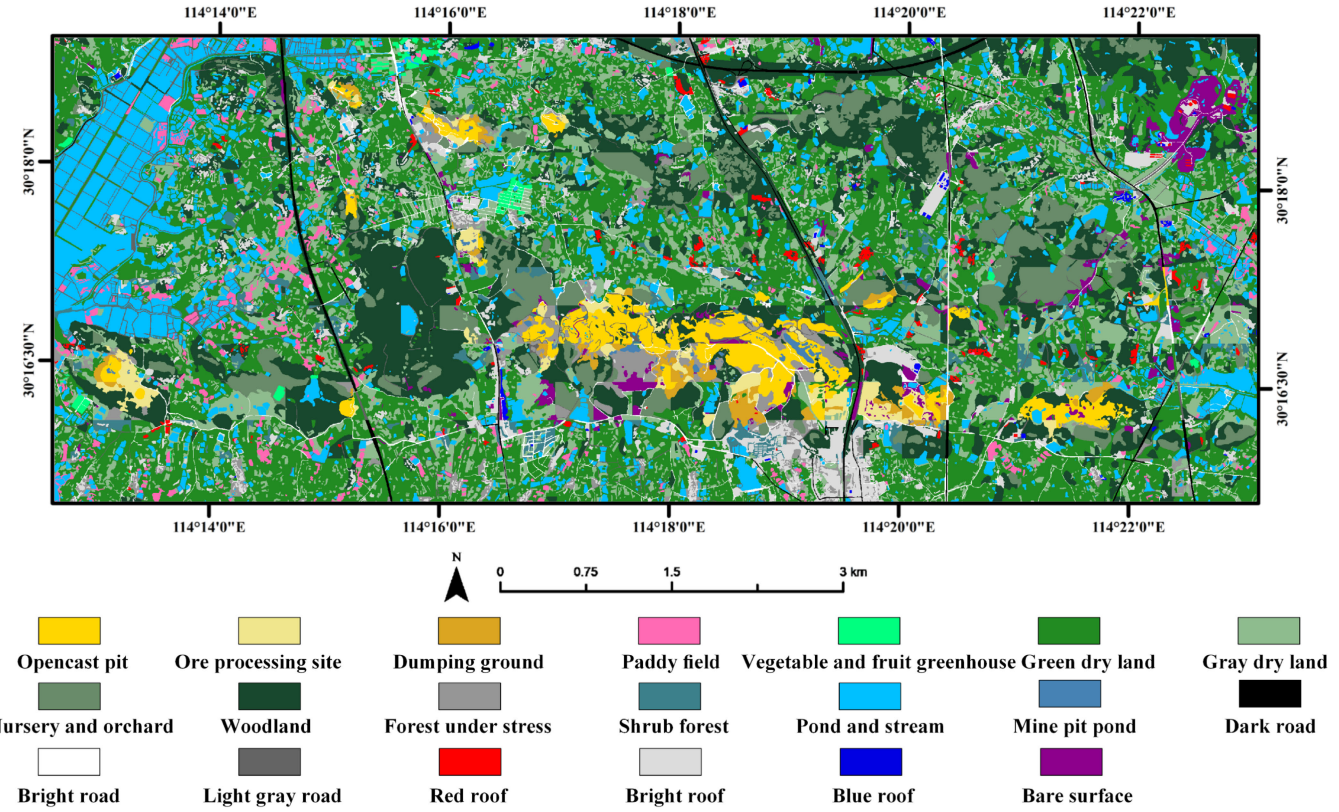


Fig. 6. Results of visual interpretation of the study area.

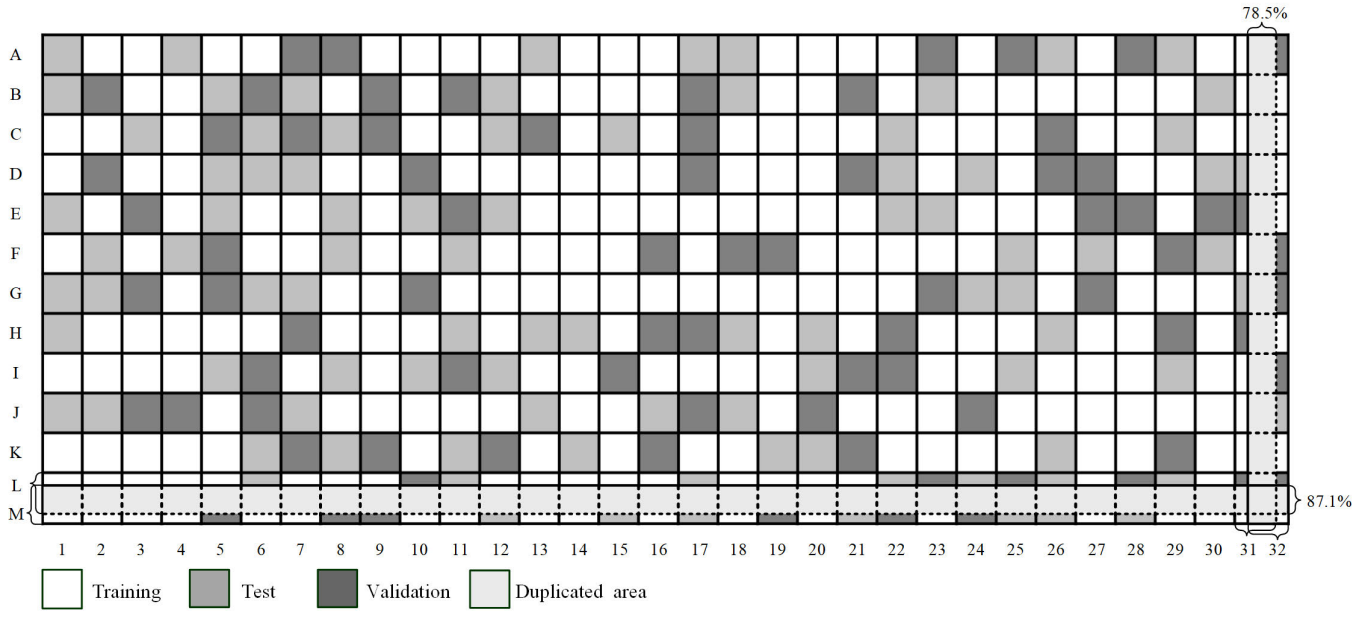


Fig. 7. Dataset distribution in the study area.

areas *M* and *L*, and a 78.5% area duplication existed between areas 31 and 32.

We randomly sampled the training, test, and validation sets from areas A to M and ensured that the three datasets had samples in each area so that the data were evenly distributed in space. Moreover, to maintain consistency in the class distribution between datasets as far as possible, the dataset was manually fine-tuned following random sampling to achieve a distribution balance among the classes in the training, test, and validation sets. Finally, there were 247 images in the training set, 91 in the test set, and 78 inthe validation set, at a ratio of 6:2:2. Fig 7 depicts the dataset distribution in the study area. Table II provides the details of the proposed LCMA dataset. Moreover,

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TABLE II
CLASS DISTRIBUTIONS IN TRAINING, TEST, AND VALIDATION SETS IN THE STUDY AREA

| No. | Fine land cover type | Class distribution (%) | | |
|---|---|---|---|---|
| | | Training | Test | Validation |
| 1 | Opencast pit | 2.58 | 2.85 | 2.03 |
| 2 | Ore processing site | 0.94 | 0.47 | 0.79 |
| 3 | Dumping ground | 0.92 | 0.92 | 0.54 |
| 4 | Paddy field | 1.83 | 1.95 | 2.31 |
| 5 | Vegetable and fruit greenhouse | 0.41 | 0.52 | 0.23 |
| 6 | Green dry land | 29.67 | 28.10 | 28.63 |
| 7 | Gray dry land | 16.36 | 12.38 | 17.94 |
| 8 | Nursery and orchard | 5.83 | 5.15 | 5.08 |
| 9 | Woodland | 16.66 | 18.63 | 15.76 |
| 10 | Forest under stress | 2.34 | 2.57 | 1.57 |
| 11 | Shrub forest | 1.55 | 1.62 | 1.59 |
| 12 | Pond and stream | 11.01 | 12.37 | 12.60 |
| 13 | Mine pit pond | 0.09 | 0.27 | 0.10 |
| 14 | Dark road | 0.65 | 0.65 | 1.02 |
| 15 | Bright road | 1.62 | 1.50 | 1.54 |
| 16 | Light gray road | 2.03 | 2.35 | 2.28 |
| 17 | Red roof | 0.51 | 0.63 | 0.43 |
| 18 | Bright roof | 3.51 | 4.72 | 4.43 |
| 19 | Blue roof | 0.18 | 0.22 | 0.31 |
| 20 | Bare surface | 1.32 | 2.14 | 0.83 |

TABLE III
NORMALIZATION STANDARDS OF PROPOSED LCMA DATASET IN THE STUDY AREA

| | First channel | Second channel | Third channel | Fourth channel | Fifth channel |
|---|---|---|---|---|---|
| Max. value | 122 | 1375 | 1583 | 1267 | 2612 |
| Mean value (after dividing by max. value) | 0.209 | 0.394 | 0.380 | 0.344 | 0.398 |
| Standard deviation | 0.141 | 0.027 | 0.032 | 0.046 | 0.108 |

Table III lists the normalization standards of the proposed dataset. This dataset can be downloaded from the following website: https://drive.google.com/drive/folders/1GeuAU97Z-_Eaj7r0QTI2cjwX0zR3nmEQ?usp=sharing.

## IV. EXPERIMENTAL RESULTS

In this section, we compare the results of different structures in the proposed model, including the various backbones and modules. Then, we present several comparative experiments conducted using other deep learning models. U-Net is a classical and commonly used semantic segmentation model. PSPNet uses a pyramid pooling module to aggregate the global context information. DeepLab v3+ uses dilated convolution and atrous spatial pyramid pooling to expand the receptive field and capture contextual information. UNet++ focuses on multiscale features. Attention U-Net uses an attention mechanism to weaken the background features. We selected these models, along with recent models, namely, ResUNet, HRNet, Swin UNet, and DE-DCGCN-EE, for comparisons of the effectiveness of the EE and GCN modules in our proposed method.

All experiments in the study were conducted for five rounds, and the mean value and standard deviation were used for the comparisons.

The environment and framework used were Python v3.6 and PyTorch 1.1.0, respectively. The hardware configuration had 64 GB of memory, and the CPU was an Intel[1] Xeon[1] Silver 4210 and RTX2080ti GPU with 11 GB of memory. The batch size was 8. Furthermore, cross-entropy was used as the loss function, whereas ReLU was used as the activation function.

### A. Evaluation Metrics

We implemented a pixel-based semantic segmentation task for the experiments. The proposed model and other mainstream deep learning models were evaluated using the overall accuracy (OA), F1 score, Intersection over Union (IoU), mean pixel intersection-over-union (mIOU), macro-F1 (average value of F1 scores), floating-point operations (FLOPs), and the number of parameters (Param). The evaluation metrics are

[1]Registered trademark.

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TABLE IV
QUANTITATIVE COMPARISON OF DIFFERENT BACKBONES AND MODULES USING THE PROPOSED LCMA DATASET

| Class | Different Backbones | | | | | | | | Different Modules | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | EG-ResNet-50 | | EG-VGG16 | | EG-UNet++ | | EG-UNet | | EE&U-Net | | GCN&U-Net | |
| | IOU | F1 | IOU | F1 | IOU | F1 | IOU | F1 | IOU | F1 | IOU | F1 |
| Opencast pit | 37.18 ±2.54 | 54.16 ±2.69 | 43.48 ±3.11 | 60.55 ±3.01 | 42.50 ±2.39 | **63.26** ±8.15 | 42.32 ±4.48 | 59.36 ±4.48 | **48.10** ±7.78 | 60.36 ±3.53 | 44.85 ±2.36 | 61.89 ±2.27 |
| Ore processing site | 3.51 ±3.55 | 6.60 ±6.54 | 3.46 ±2.47 | 6.61 ±4.64 | 0 | 0 | **6.35** ±1.51 | **11.92** ±2.65 | 3.93 ±2.24 | 7.49 ±4.13 | 2.70 ±2.95 | 4.74 ±5.81 |
| Dumping ground | 14.65 ±4.38 | 25.35 ±6.74 | 8.96 ±8.17 | 15.66 ±13.3 | 0 | 0 | 12.84 ±2.61 | 22.68 ±4.18 | **15.39** ±4.99 | **26.40** ±7.56 | 11.65 ±5.57 | 20.51 ±8.95 |
| Paddy field | 53.17 ±1.69 | 69.41 ±1.44 | 55.03 ±1.35 | 70.98 ±1.13 | 53.61 ±2.83 | 68.85 ±2.32 | 54.43 ±2.09 | 70.47 ±1.77 | **55.12** ±0.57 | **71.07** ±0.47 | 54.78 ±1.37 | 70.78 ±1.14 |
| Vegetable and fruit greenhouse | 7.32 ±11.7 | 12.02 ±18.6 | 5.04 ±11.2 | 8.06 ±17.9 | 0 | 0 | **28.32** ±6.24 | **43.85** ±7.51 | 5.10 ±11.3 | 8.16 ±17.9 | 17.29 ±16.4 | 26.69 ±24.9 |
| Green dry land | 49.45 ±1.04 | 66.17 ±0.93 | 51.74 ±1.47 | 68.18 ±1.27 | **51.82** ±1.26 | **69.50** ±2.55 | 50.90 ±0.62 | 67.46 ±0.55 | 50.89 ±1.58 | 67.44 ±1.39 | 51.31 ±1.35 | 67.81 ±1.18 |
| Gray dry land | 34.56 ±1.83 | 51.35 ±2.01 | **36.68** ±1.24 | 53.66 ±1.32 | 35.97 ±1.04 | **55.26** ±5.53 | 35.92 ±1.06 | 52.85 ±1.15 | 36.11 ±1.71 | 53.04 ±1.86 | 36.11 ±0.63 | 53.06 ±0.68 |
| Nursery and orchard | 39.72 ±2.07 | 56.83 ±2.14 | 40.47 ±1.97 | 57.60 ±2.03 | 40.79 ±2.53 | 57.63 ±2.20 | **41.26** ±1.66 | **58.40** ±1.68 | 40.04 ±1.72 | 57.16 ±1.77 | 40.94 ±1.10 | 58.09 ±1.10 |
| Woodland | 37.70 ±2.29 | 54.73 ±2.41 | 40.69 ±1.88 | 57.83 ±1.92 | **41.76** ±1.16 | **58.07** ±2.32 | 39.55 ±1.00 | 56.68 ±1.02 | 37.33 ±1.69 | 54.34 ±1.81 | 39.68 ±1.93 | 56.79 ±1.99 |
| Forest under stress | 5.85 ±2.17 | 10.98 ±3.89 | 12.51 ±3.27 | 22.11 ±5.34 | 5.21 ±5.34 | 9.35 ±9.68 | 12.47 ±2.10 | 22.12 ±3.31 | **14.52** ±2.18 | **25.30** ±3.35 | 10.88 ±2.58 | 19.56 ±4.15 |
| Shrub forest | 0 | 0 | 0.05 ±0.1 | 0.10 ±0.21 | 0 | 0 | **0.70** ±0.93 | **1.37** ±1.81 | 0.29 ±0.64 | 0.56 ±1.26 | 0.33 ±0.45 | 0.65 ±0.90 |
| Pond and stream | 78.37 ±1.23 | 87.87 ±0.78 | 78.68 ±0.44 | 88.07 ±0.28 | **79.18** ±0.23 | **88.52** ±0.25 | 78.03 ±0.82 | 87.66 ±0.52 | 78.54 ±0.53 | 87.98 ±0.33 | 78.39 ±0.94 | 87.89 ±0.59 |
| Mine pit pond | 1.55 ±1.39 | 3.02 ±2.69 | 0.05 ±0.12 | 0.11 ±0.24 | 0 | 0 | **2.74** ±0.97 | **5.33** ±1.82 | 0.39 ±0.72 | 0.77 ±6.04 | 2.00 ±3.87 | 3.72 ±7.09 |
| Dark road | 40.37 ±9.63 | 56.97 ±9.92 | 21.09 ±7.35 | 57.92 ±7.78 | 38.19 ±11.2 | 52.00 ±12.2 | **47.43** ±3.12 | **64.30** ±2.90 | 45.46 ±6.58 | 62.28 ±6.04 | 46.24 ±0.91 | 63.24 ±0.85 |
| Bright road | 26.69 ±2.87 | 42.07 ±3.54 | 25.36 ±2.91 | 40.39 ±3.71 | 27.93 ±1.18 | 44.95 ±3.08 | 28.68 ±1.41 | 44.56 ±1.71 | 24.90 ±2.33 | 39.82 ±2.99 | 27.38 ±1.53 | 42.97 ±1.89 |
| Light gray road | 13.59 ±2.30 | 23.87 ±3.61 | 13.37 ±1.85 | 23.54 ±2.89 | 10.73 ±2.36 | 17.86 ±5.13 | **16.26** ±1.04 | **27.96** ±1.53 | 12.49 ±3.62 | 22.07 ±5.65 | 14.60 ±2.59 | 25.40 ±3.99 |
| Red roof | 0.72 ±0.53 | 1.43 ±1.05 | 0.36 ±0.49 | 0.72 ±0.97 | 0 | 0 | **5.39** ±3.05 | **10.11** ±5.59 | 1.26 ±1.60 | 2.44 ±3.11 | 1.25 ±2.13 | 2.41 ±4.05 |
| Bright roof | 43.96 ±3.88 | 60.99 ±3.78 | 46.24 ±3.68 | 63.17 ±3.46 | **49.48** ±3.15 | **67.08** ±4.57 | 47.83 ±2.55 | 64.68 ±2.37 | 47.44 ±1.73 | 64.34 ±1.59 | 47.97 ±1.48 | 64.83 ±1.35 |
| Blue roof | 0.43 ±0.36 | 0.85 ±0.71 | 0.04 ±0.08 | 0.07 ±0.16 | 0 | 0 | **0.87** ±1.33 | **1.69** ±2.59 | 0 | 0 | 0.30 ±0.22 | 0.60 ±1.08 |
| Bare surface | 31.88 ±3.99 | 48.23 ±4.58 | 31.85 ±1.95 | 48.29 ±2.25 | 16.90 ±15.7 | 23.62 ±22.6 | 32.98 ±6.28 | 49.33 ±7.41 | **35.55** ±4.62 | **52.31** ±4.92 | 31.16 ±4.32 | 47.38 ±4.98 |
| OA | 59.87±1.29 | | 61.68±0.89 | | **61.90**±0.66 | | 61.41±0.44 | | 61.19±0.52 | | 61.59±0.83 | |
| mIOU | 26.03±2.05 | | 26.76±1.72 | | 24.70±1.61 | | **29.26**±0.47 | | 27.40±1.43 | | 27.99±1.06 | |
| Macro-F1 | 36.65±2.62 | | 37.18±2.43 | | 33.80±2.24 | | **41.14**±0.55 | | 38.17±2.11 | | 38.95±1.94 | |

defined as follows:

$$\text{Overall accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}} \tag{7}$$

$$\text{F1 score} = \frac{2\text{TP}}{2\text{TP} + \text{FP} + \text{FN}} \tag{8}$$

$$\text{IOU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \tag{9}$$

where true positive (TP) is the number of pixels that are correctly extracted as a road, false positive (FP) is the number of other object pixels that are extracted as a road, true negative (TN) is the number of pixels that are correctly extracted as the background, and false negative (FN) is the number of road pixels extracted as the background.

### B. Comparison of Different Structures in EG-UNet

This section examines the use of different backbone networks and module combinations in the proposed model.

Table IV lists the evaluation metrics for the results of the proposed model with different backbones and modules when using the test set of the proposed model. EG represents the use of both the EE and GCN modules. The meanings of classes 1–20 are provided in Tables I and II. We used three colors to label the classes: red indicates classes with very few samples (the number of samples is less than 1%), yellow indicates classes with few samples (the number of samples is between 1% and 3%), and black indicates classes with a large number of samples (more than 3%). We selected several

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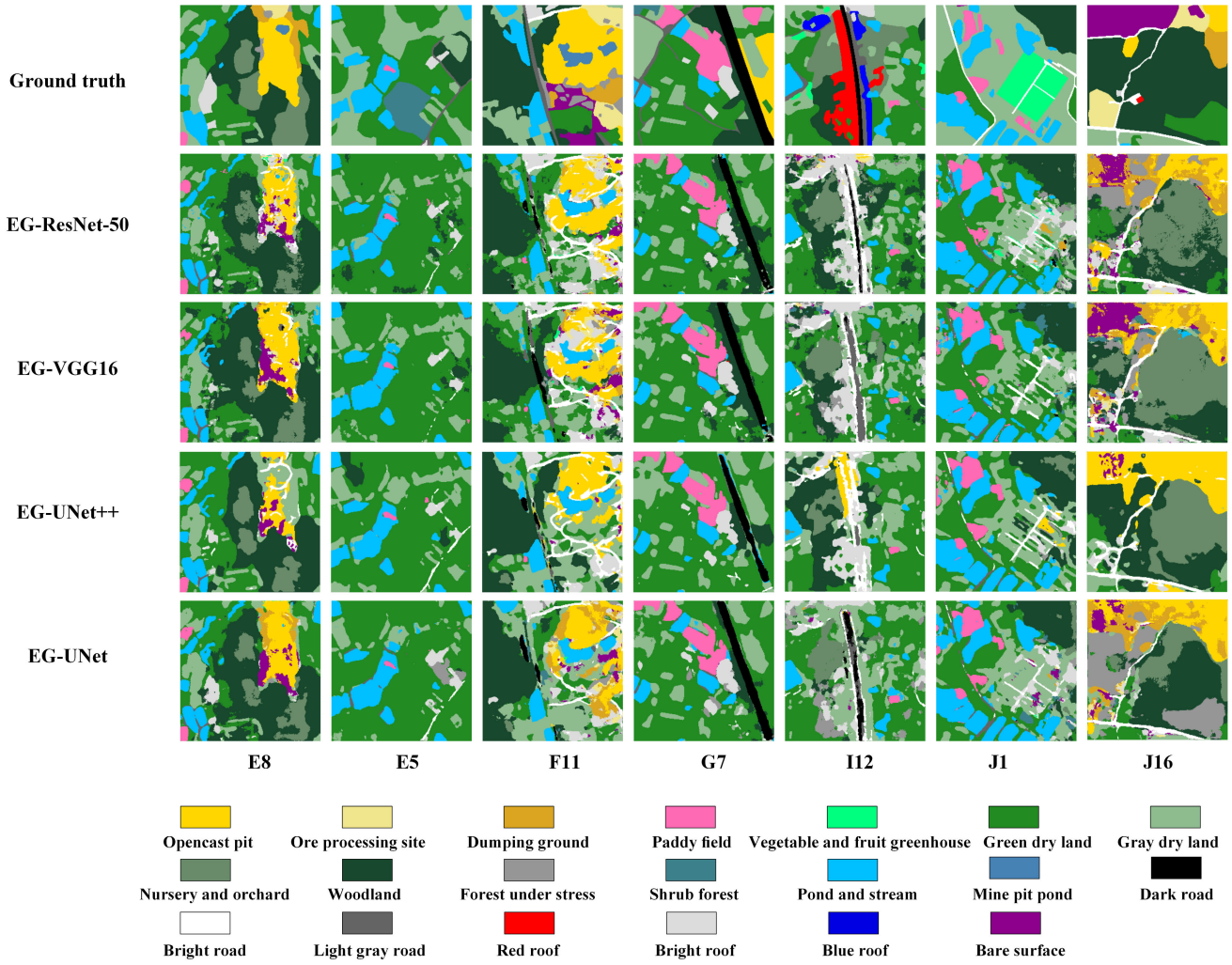


Fig. 8. Visual comparison of different backbones using representative images (from E8 to J16) in the study area.

classical networks as the basic framework (U-Net, UNet++, VGG16, and ResNet-50). EG-UNet yielded the best results in terms of the mIOU and macro-F1, especially for the classes with few and very few samples. The results in Table IV verify that the EE and GCN modules in the proposed model are beneficial.

Figs. 8 and 9 depict the performances of the different module combinations, as detailed in Table IV. Seven representative images were used for comparison. Area E8 contained the most types of classes; E5 contained shrub forest and light gray road (classes with few samples); F11 contained ore processing site, dumping ground, and mine pit pond (classes with very few samples); G7 contained paddy field, opencast pit, dry green land, and woodland (classes with few samples); I12 contained dark road, blue roof, and red roof (classes with very few samples); J1 contained vegetable and fruit greenhouse (classes with very few samples); and J16 contained bright road and bare surface (classes with few samples). Although a single area contained many categories, differences in recognition of certain categories could be observed when using different strategies, in particular, dark road in I12, opencast pit and dumping ground in E8, bare surface and woodland in J16, and bright road in J1. Figs. 7 and 8 indicate that EG-UNet exhibited good performance in classes with few samples, which could be attributed to the EE and GCN modules. As can be observed from Fig. 8, the results were improved by including the two modules. For example, in E8, the classification accuracy of EE&UNet was better than that of U-Net, and for woodland in E5, GCN&U-Net was better than that of U-Net.

### C. Model Comparison in the LCMA Dataset

Table V summarizes the evaluation metrics for the results of the proposed model and comparison models when using the test set of the proposed dataset. Among these methods, the proposed model exhibited the best performance, with an mIOU of 29.26%, a macro-F1 of 41.14%, and an OA of 61.41%. Compared to the unmodified U-Net, the mIOU of EG-UNet increased by 2.26%, and the macro-F1 score increased by 3.32%. EG-UNet yielded the best results for six classes with few samples (ore processing site, bright road, light gray road, red roof, blue roof, and bare surface), accounting for 42.9% of the total number of classes.

Fig. 10 depicts the performance of the seven methods, as outlined in Table V. DeepLab v3+ and PSPNet performed

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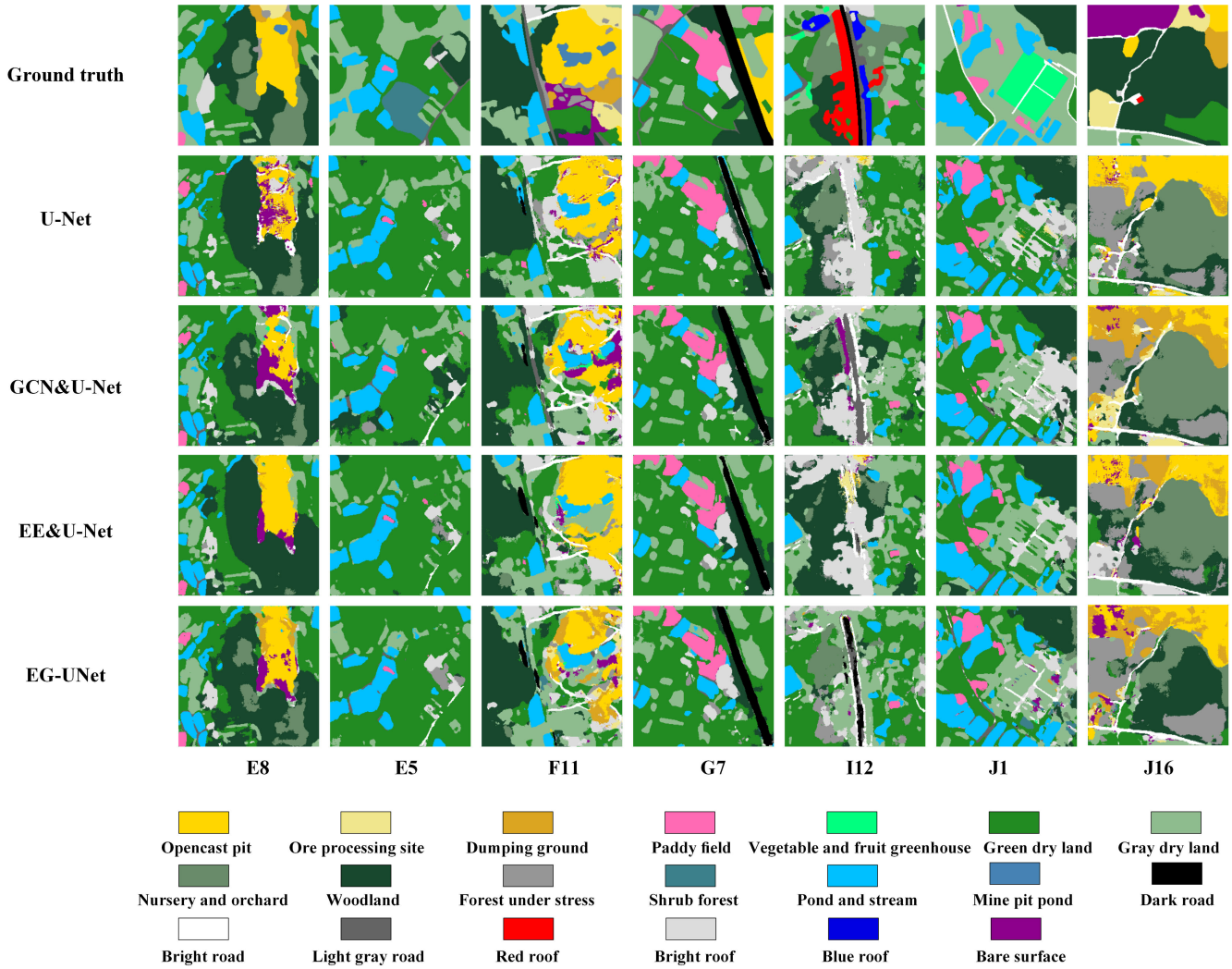


Fig. 9. Visual comparison of different modules of EG-UNet using representative images (from E8 to J16) in the study area.

poorly, so their visual results are not included in the figure. HRNet, ResUNet, and EG-UNet exhibited superior performance compared to the other models. EG-UNet yielded better performance for dumping ground in E8 and F11, paddy field in E5, dark road in G7 and I12, and bare surface and woodland in J16. However, HRNet was superior for dumping ground in J16, and EG-UNet misclassified woodland in E5.

Although the comprehensive performance of EG-UNet was the best, many categories could not be accurately classified, such as the mine pit pond in F11 and the shrub forest in E5.

### *D. Model Comparison in GID*

To evaluate our proposed model's general ability, we applied a fine land-cover classification set of GID as a benchmark, which has superiorities over the existing land-cover dataset because of its extensive coverage, wide distribution, and high spatial resolution. GID was constructed with a Gaofen-2 satellite image, and Chinese Land Use Classification Criteria (GB/T21010-2017) were used to determine a hierarchical category system. It comprises 16 subcategories: garden land, paddy field, irrigated land, arbor forest, dry cropland, natural meadow, shrub land, traffic land, artificial meadow, urban residential, rural residential, industrial land, river, lake, pond, and others. There are ten imageries in the dataset. Each image provides a spatial dimension of 6800 × 7200 pixels covering a geographic area of 506 km$^2$. Fig. 11 shows samples from the dataset. We select six images as training sets, two as testing, and two as validation sets. Each image was divided into small batches with sizes of 224 × 224 pixels. Table VI presents the batch number of the training, validation, and testing datasets. Table VII provides the class distribution of the GID dataset. In the test dataset, we calculate the class distribution of each original image (A and B).

U-Net, Attention, DeepLab v3+, SwinUNet, and DE-DCGCN-EE are used to compare performance with EG-UNet. Tables VIII and IX summarize the evaluation metrics of each original image in the test dataset.

In Tables VIII and IX, we also use red, yellow, and black to label the classes, the same as in Table IV. According to Table VIII, there are 11 classes in image A. Among these methods, DE-DCGCN-EE performs best, with an mIOU of 32.89%, a macro-F1 of 39.53%, and an OA of 73.58%. The

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TABLE V
PERFORMANCE COMPARISON OF DIFFERENT MODELS ON THE PROPOSED LCMA DATASET

| Class | U-Net | | UNet++ | | Attention U-Net | | DeepLab v3+ | | SwinUNet | | DE-DCGCN-EE | | PSPNet | | ResUNet | | HRNet | | EG-UNet | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | IOU | F1 | IOU | F1 | IOU | F1 | IOU | F1 | IOU | F1 | IOU | F1 | IOU | F1 | IOU | F1 | IOU | F1 | IOU | F1 |
| Opencast pit | 42.48 | 59.48 | 42.64 | 59.78 | 40.90 | 57.99 | 33.25 | 49.56 | 45.41 | 62.42 | 34.04 | 50.59 | 37.46 | 54.48 | **43.02** | **60.15** | 41.58 | 58.72 | 42.32 | 59.36 |
| | ±5.10 | ±5.19 | ±1.40 | ±1.37 | ±3.51 | ±3.53 | ±7.18 | ±8.13 | ±2.72 | ±2.58 | ±5.48 | ±6.03 | ±1.92 | ±2.02 | ±1.85 | ±1.80 | ±2.11 | ±2.10 | ±4.48 | ±4.48 |
| Ore | 4.28 | 8.10 | 0 | 0 | 0.02 | 0.04 | 1.89 | 3.70 | 0.11 | 0.22 | 1.49 | 2.81 | 0.73 | 1.44 | 3.48 | 6.57 | 2.85 | 5.51 | **6.35** | **11.92** |
| processing | ±2.89 | ±5.20 | | | ±0.04 | ±0.09 | ±1.09 | ±2.08 | ±0.09 | ±0.20 | ±3.01 | ±5.62 | ±0.62 | ±1.23 | ±3.28 | ±6.14 | ±1.20 | ±2.28 | ±1.51 | ±2.65 |
| site | | | | | | | | | | | | | | | | | | | | |
| Dumping | **15.43** | **26.53** | 0 | 0 | 0.91 | 1.74 | 4.42 | 8.39 | 2.95 | 5.68 | 3.79 | 7.23 | 1.99 | 3.90 | 2.78 | 5.20 | 8.14 | 14.96 | 12.84 | 22.68 |
| ground | ±4.36 | ±6.89 | | | ±2.03 | ±3.89 | ±2.46 | ±4.49 | ±1.74 | ±3.30 | ±2.25 | ±4.24 | ±0.89 | ±1.73 | ±3.89 | ±7.16 | ±2.77 | ±4.78 | ±2.61 | ±4.18 |
| Paddy field | 54.54 | 70.58 | 54.86 | 70.84 | **55.03** | **70.99** | 49.67 | 66.37 | 39.26 | 56.36 | 51.04 | 67.58 | 48.91 | 65.67 | 53.54 | 69.72 | 54.05 | 70.14 | 54.43 | 70.47 |
| | ±1.23 | ±1.04 | ±1.42 | ±1.19 | ±0.65 | ±0.54 | ±0.83 | ±0.74 | ±1.77 | ±1.81 | ±0.97 | ±0.85 | ±1.78 | ±1.62 | ±2.11 | ±1.78 | ±2.58 | ±2.21 | ±2.09 | ±1.77 |
| Vegetable | 5.79 | 10.11 | 0 | 0 | 0 | 0 | 0.06 | 0.12 | 0.01 | 0.02 | 0 | 0 | 0.84 | 1.62 | 0 | 0 | **33.57** | **50.25** | 28.32 | 43.85 |
| and fruit | ±7.96 | ±13.9 | | | | | ±0.14 | ±0.27 | ±0.03 | ±0.05 | | | ±1.79 | ±3.44 | | | ±1.85 | ±2.09 | ±6.24 | ±7.51 |
| greenhouse | | | | | | | | | | | | | | | | | | | | |
| Green dry | 51.45 | 67.94 | 51.52 | 68.00 | **51.85** | **68.29** | 45.89 | 62.91 | 47.43 | 64.34 | 47.68 | 64.54 | 46.05 | 63.05 | 49.79 | 66.48 | 50.01 | 66.68 | 50.90 | 67.46 |
| land | ±0.59 | ±0.52 | ±1.01 | ±0.88 | ±0.52 | ±0.45 | ±0.68 | ±0.64 | ±0.42 | ±0.39 | ±2.41 | ±2.22 | ±1.02 | ±0.95 | ±0.51 | ±0.45 | ±0.65 | ±0.58 | ±0.62 | ±0.55 |
| Gray dry | 36.25 | 53.21 | 35.88 | 52.80 | 35.15 | 52.00 | 31.46 | 47.86 | 32.43 | 48.97 | 29.39 | 45.36 | 32.44 | 48.96 | 37.04 | 54.04 | **37.72** | **54.77** | 35.92 | 52.85 |
| land | ±0.90 | ±0.97 | ±0.87 | ±0.94 | ±1.58 | ±1.74 | ±0.59 | ±0.68 | ±0.54 | ±0.62 | ±3.04 | ±3.65 | ±2.14 | ±2.45 | ±1.10 | ±1.17 | ±1.01 | ±1.05 | ±1.06 | ±1.15 |
| Nursery and | 39.06 | 56.16 | **41.69** | **58.85** | 40.82 | 57.96 | 36.16 | 53.09 | 37.08 | 54.09 | 32.08 | 48.03 | 37.34 | 54.16 | 29.60 | 43.20 | 40.04 | 57.54 | 41.26 | 58.40 |
| orchard | ±1.12 | ±1.16 | ±0.73 | ±0.73 | ±1.99 | ±2.01 | ±1.93 | ±2.10 | ±1.23 | ±1.29 | ±8.74 | ±10.2 | ±5.88 | ±6.43 | ±16.6 | ±24.2 | ±0.97 | ±0.99 | ±1.66 | ±1.68 |
| Woodland | 36.81 | 53.81 | **40.68** | **57.80** | 40.22 | 57.36 | 30.53 | 46.75 | 34.47 | 51.24 | 35.47 | 52.17 | 35.92 | 52.78 | 39.40 | 56.53 | 38.90 | 55.99 | 39.55 | 56.68 |
| | ±0.61 | ±0.65 | ±2.21 | ±2.26 | ±1.18 | ±1.21 | ±1.70 | ±2.01 | ±2.24 | ±2.49 | ±5.52 | ±5.23 | ±3.42 | ±3.72 | ±0.82 | ±0.85 | ±1.96 | ±2.02 | ±1.00 | ±1.02 |
| Forest under | 13.61 | 23.94 | 3.74 | 7.03 | 3.76 | 7.24 | 10.89 | 19.55 | 7.14± | 13.32 | 7.04 | 13.06 | 9.29 | 16.75 | 11.01 | 19.62 | **15.47** | **26.76** | 12.47 | 22.12 |
| stress | ±1.42 | ±2.21 | ±3.69 | ±6.82 | ±1.06 | ±1.97 | ±2.64 | ±4.33 | 0.35 | ±0.61 | ±2.70 | ±4.76 | ±4.57 | ±7.67 | ±4.21 | ±7.04 | ±1.63 | ±2.41 | ±2.10 | ±3.31 |
| Shrub forest | 0.21 | 0.42 | 0 | 0 | 0 | 0 | 0.06 | 0.12 | 0 | 0 | 0 | 0 | **0.82** | **1.62** | 0 | 0 | 0.43 | 0.86 | 0.70 | 1.37 |
| | ±0.47 | ±0.93 | | | | | ±0.06 | ±0.11 | | | | | ±1.06 | ±2.08 | | | ±0.26 | ±0.51 | ±0.93 | ±1.81 |
| Pond and | 78.50 | 87.95 | **79.06** | **88.30** | 78.70 | 88.08 | 72.59 | 84.10 | 77.50 | 87.32 | 76.06 | 86.39 | 72.19 | 83.85 | 78.13 | 87.72 | 79.02 | 88.28 | 78.03 | 87.66 |
| stream | ±0.51 | ±0.32 | ±1.13 | ±0.71 | ±0.36 | ±0.22 | ±2.41 | ±1.61 | ±1.06 | ±0.68 | ±2.20 | ±1.40 | ±0.92 | ±0.62 | ±0.95 | ±0.60 | ±0.18 | ±0.11 | ±0.82 | ±0.52 |
| Mine pit | 0.06 | 0.11 | 0 | 0 | 0 | 0 | 0.42 | 0.84 | 0.21 | 0.41 | 0 | 0 | 2.43 | 4.58 | 0 | 0 | **6.98** | **12.97** | 2.74 | 5.33 |
| pond | ±0.13 | ±0.25 | | | | | ±0.64 | ±1.26 | ±0.22 | ±0.44 | | | ±3.33 | ±6.28 | | | ±2.51 | ±4.51 | ±0.97 | ±1.82 |
| Dark road | 44.65 | 61.69 | 21.21 | 30.56 | 9.68 | 16.60 | 13.67 | 22.99 | 24.59 | 39.28 | 11.17 | 19.53 | 39.58 | 56.57 | 19.99 | 26.65 | **53.00** | **69.28** | 47.43 | 64.30 |
| | ±3.12 | ±3.00 | ±22.1 | ±30.5 | ±9.18 | ±15.6 | ±10.2 | ±14.8 | ±4.83 | ±6.28 | ±7.04 | ±11.2 | ±4.88 | ±5.08 | ±27.4 | ±36.5 | ±1.07 | ±0.91 | ±3.12 | ±2.90 |
| Bright road | 24.70 | 39.59 | 27.27 | 42.82 | 24.17 | 38.89 | 15.21 | 26.15 | 7.66 | 14.16 | 15.27 | 26.45 | 9.54 | 17.26 | 25.31 | 40.04 | 27.91 | 43.57 | **28.68** | **44.56** |
| | ±1.61 | ±2.08 | ±1.91 | ±2.38 | ±2.27 | ±2.94 | ±5.02 | ±7.52 | ±2.35 | ±4.03 | ±2.09 | ±3.20 | ±3.59 | ±5.94 | ±6.40 | ±8.66 | ±3.06 | ±3.86 | ±1.41 | ±1.71 |
| Light gray | 13.62 | 23.93 | 9.75 | 17.58 | 10.06 | 18.27 | 8.17 | 15.03 | 5.19 | 9.85 | 8.00 | 14.60 | 4.11 | 7.80 | 14.58 | 25.28 | 15.91 | 27.45 | **16.26** | **27.96** |
| road | ±1.89 | ±2.93 | ±3.90 | ±6.62 | ±0.66 | ±1.10 | ±2.48 | ±4.34 | ±1.38 | ±2.48 | ±4.20 | ±7.20 | ±2.53 | ±4.60 | ±3.91 | ±5.96 | ±0.99 | ±1.47 | ±1.04 | ±1.53 |
| Red roof | 0.66 | 1.29 | 0 | 0 | 0 | 0 | 1.42 | 2.76 | 0 | 0 | 0 | 0 | 0.46 | 0.91 | 0 | 0 | 1.05 | 2.07 | **5.39** | **10.11** |
| | ±1.32 | ±2.56 | | | | | ±1.44 | ±2.79 | | | | | ±0.70 | ±1.39 | | | ±0.98 | ±1.91 | ±3.05 | ±5.59 |
| Bright roof | 46.68 | 63.64 | 44.22 | 61.24 | 39.86 | 56.96 | 35.25 | 52.07 | 33.77 | 50.46 | 27.78 | 43.12 | 42.71 | 59.82 | 46.41 | 63.39 | 46.94 | 63.88 | **47.83** | **64.68** |
| | ±1.30 | ±1.20 | ±3.71 | ±3.68 | ±2.68 | ±2.75 | ±2.88 | ±3.10 | ±2.14 | ±2.37 | ±6.71 | ±8.78 | ±2.37 | ±2.33 | ±1.29 | ±1.19 | ±1.24 | ±1.15 | ±2.55 | ±2.37 |
| Blue roof | 0.63 | 1.21 | 0 | 0 | 0 | 0 | 0.30 | 0.59 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0.12 | 0.23 | **0.87** | **1.69** |
| | ±1.39 | ±2.69 | | | | | ±0.46 | ±0.92 | | | | | | | | | ±0.24 | ±0.47 | ±1.33 | ±2.59 |
| Bare surface | 30.52 | 46.62 | 8.62 | 14.12 | 26.71 | 39.94 | 9.53 | 16.69 | 25.01 | 39.80 | 22.66 | 36.43 | 11.66 | 20.37 | 30.95 | 46.97 | 16.42 | 27.99 | **32.98** | **49.33** |
| | ±4.55 | ±5.38 | ±12.5 | ±19.2 | ±15.7 | ±22.5 | ±7.71 | ±12.6 | ±5.07 | ±6.75 | ±7.73 | ±10.2 | ±6.76 | ±10.7 | ±6.39 | ±7.71 | ±4.59 | ±6.74 | ±6.28 | ±7.41 |
| OA | 61.00±0.61 | | 61.25±0.76 | | 60.93±0.67 | | 55.01±0.85 | | 57.11±0.59 | | 56.17±1.35 | | 56.68±0.83 | | 60.45±0.48 | | 61.00±0.46 | | **61.41**±0.44 | |
| mIOU | 27.00±0.60 | | 23.06±1.99 | | 22.89±1.56 | | 20.04±1.26 | | 21.01±0.8 | | 20.15±0.57 | | 21.72±1.13 | | 24.25±2.00 | | 28.52±0.56 | | **29.26**±0.47 | |
| Macro-F1 | 37.82±0.99 | | 31.49±2.76 | | 31.62±2.26 | | 28.98±1.65 | | 29.90±1.04 | | 28.89±0.96 | | 30.78±1.72 | | 33.58±2.79 | | 39.90±0.78 | | **41.14**±0.55 | |
| FLOPs | 21.02G | | 42.64G | | 21.42G | | 8.55G | | 11.92G | | 148.22G | | 19.17G | | 31.09G | | 8.75G | | 22.15G | |
| Param | 31.04M | | 34.92M | | 31.39M | | 59.35M | | 27.17M | | 14.91M | | 48.70M | | 13.05M | | 29.51M | | 42.56M | |

proposed model is about only 0.5% lower than DE-DCGCN-EE on the metrics. Furthermore, it yields the best results for two classes with few samples (rural residential and traffic land) and needs less computation. In Table IX, the proposed model exhibited the best performance, with an mIOU of 29.63%, a macro-F1 of 38.84%, and an OA of 56.43%. Compared to the

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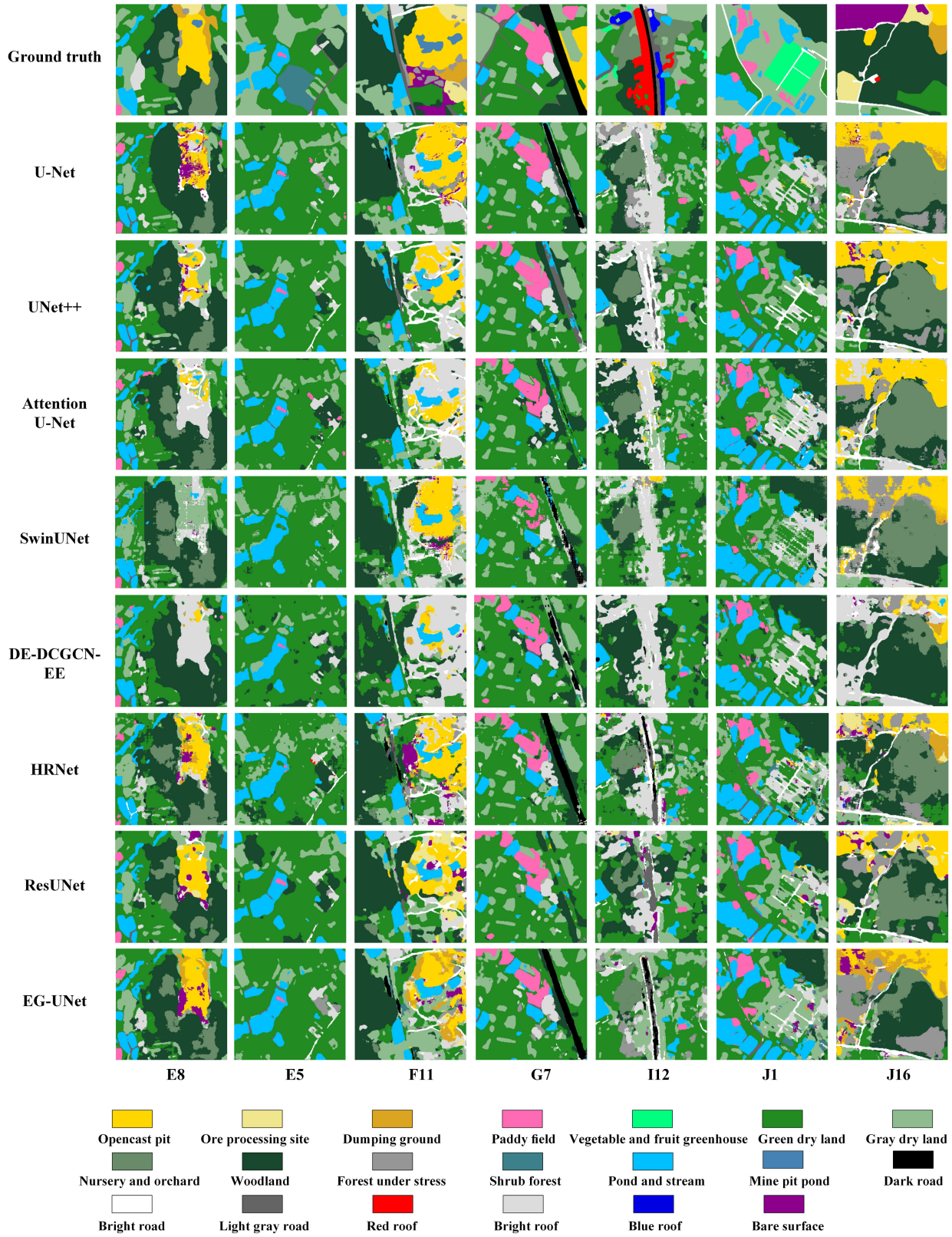


Fig. 10. Visual comparison of different models using representative images (from E8 to J16) in the study area.

second-best model (DE-DCGCN-EE), the mIOU of EG-UNet increased by 1.77%, and the macro-F1 score increased by 2.32%. EG-UNet yielded the best results for two classes with few samples (traffic land and river).

Fig. 12 depicts the performance of the six methods, as outlined in Tables VIII and IX. DE-DCGCN-EE and EG-UNet exhibited superior performance compared to the other models. EG-UNet yielded better performance for traffic land in A and

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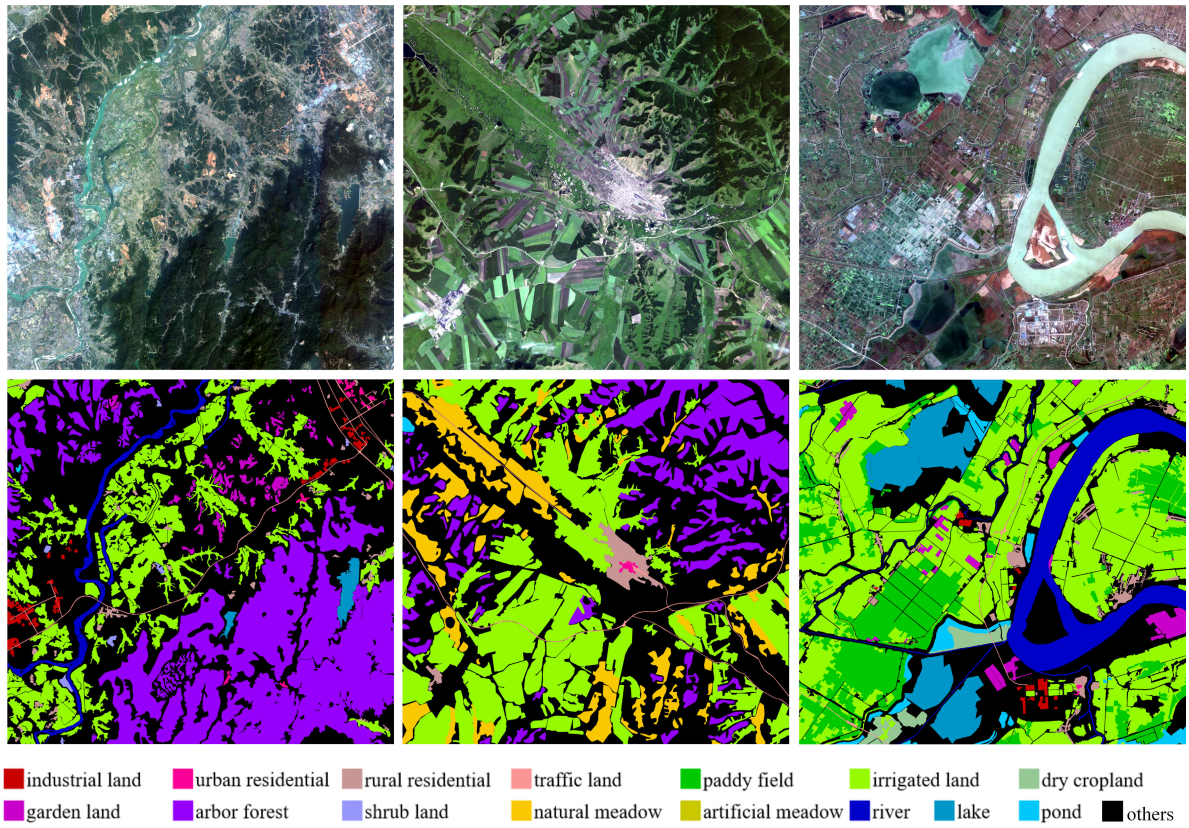


Fig. 11. Example image and label map from the fine land-cover classification set-GID [73].

TABLE VI
BATCH NUMBER OF GID USED IN THIS STUDY

| Image Size | Training Set | Validation Set | Test Set |
|---|---|---|---|
| 224 × 224 | 6138 | 2046 | 2046 |

B and rural residential in B. However, DE-DCGCN-EE was superior for the river in B.

Although the comprehensive performance of EG-UNet was the best, shrubland could not be accurately classified, and the accuracy of dry cropland, garden plot, artificial, meadow, and pond is low.

## V. DISCUSSION

### A. Effects of Different Module Combinations and Innovation of Proposed Module

According to Table IV, the best results were achieved using U-Net as the framework network. U-Net has fewer network layers and a lower network complexity than ResNet and VGG16. Owing to the limited data that are included in the proposed dataset, this dataset is unsuitable for complex networks. Furthermore, with the deepening of the network layers, the features of classes with few samples will continue to be weakened by multiple pooling operations. Therefore, ResNet and VGG16 did not perform well in these classes. As UNet++ uses multiscale features to improve the accuracy, it effectively classified classes with many samples (dry green land, dry gray land, woodland, pond and stream, and bright roof in Table IV). However, this strategy is unsuitable for classes with a small number of samples, and the mIOU of EG-UNet++ was 4.56% lower than that of EG-UNet.

According to the comparison with U-Net in Table V, adding the EE and GCN modules improved the classification accuracy. The EE module focuses on classes with few samples; hence, the IOU value was improved in these classes, particularly in the opencast pit, ore processing site, dumping ground, paddy field, forest under stress, and bare surface, with a superior result to that of GCN&U-Net. The GCN module focuses on obtaining long-context features; it does not target specific classes and improves the OA.

*Innovation of EE Module:* Edge extraction is a common method in image processing. Because the mining area object is irregular and small, we designed the EE module, which combines edge extraction with depth learning, to enhance the feature of mining area objects. Unlike directly adding edge information, we take edge information as a weight matrix to enhance the feature information of edge position. Due to the long boundary of irregular shape objects, this enhancement method can strengthen the features of mine objects and weaken the features of nonmining area targets (dry green land, pond, stream, woodland, and so on) to improve the segmentation accuracy of mining area targets. The experimental results also showed the effectiveness of the EE module in LCMA.

*Innovation of a Long-Range Information Module:* GCNs are good at capturing long-distance information. Due to the

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TABLE VII
CLASS DISTRIBUTIONS IN TRAINING, TEST, AND VALIDATION SETS IN GID

| No. | Fine land cover type | Class distribution (%) | | | |
|---|---|---|---|---|---|
| | | Training | Validation | Image A in Test | Image B in Test |
| 1 | Industrial land | 4.51 | 0.95 | 6.06 | 10.64 |
| 2 | Urban residential | 8.73 | 0.04 | 14.83 | 20.43 |
| 3 | Rural residential | 3.19 | 4.51 | 1.99 | 5.01 |
| 4 | Traffic land | 2.79 | 0.81 | 4.47 | 6.97 |
| 5 | Paddy field | 2.23 | 6.81 | 0 | 0 |
| 6 | Irrigated land | 19.87 | 48.71 | 0.26 | 11.78 |
| 7 | Dry cropland | 2.51 | 1.10 | 0 | 0.71 |
| 8 | Garden plot | 0.51 | 0.99 | 0.11 | 0 |
| 9 | Arbor woodland | 6.71 | 0 | 15.42 | 1.13 |
| 10 | Shrub land | 0.24 | 0 | 0.43 | 0 |
| 11 | Natural meadow | 1.76 | 0 | 0 | 0 |
| 12 | Artificial meadow | 0.84 | 0 | 0 | 0.20 |
| 13 | River | 2.45 | 4.66 | 0 | 11.50 |
| 14 | Lake | 2.12 | 3.28 | 0.59 | 0 |
| 15 | Pond | 0.35 | 9.09 | 0.74 | 0 |
| 16 | Others | 41.42 | 19.06 | 55.11 | 31.63 |

TABLE VIII
QUANTITATIVE COMPARISON OF IMAGE A IN TEST DATASET OF GID

| Class | UNet | | Attention UNet | | Deeplab v3+ | | SwinUNet | | DE-DCGCN-EE | | EG-UNet | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | IOU | F1 | IOU | F1 | IOU | F1 | IOU | F1 | IOU | F1 | IOU | F1 |
| Industrial land | 42.30 ±0.32 | 59.45 ±0.32 | 27.28 ±1.53 | 42.84 ±1.90 | 42.00 ±1.72 | 59.14 ±1.72 | 42.03 ±2.19 | 59.16 ±2.19 | **42.93** ±4.33 | **59.99** ±4.32 | 38.75 ±3.08 | 55.81 ±3.17 |
| Urban residential | 58.57 ±1.21 | 73.86 ±0.96 | 50.29 ±0.49 | 66.92 ±0.43 | 55.49 ±0.59 | 71.37 ±0.49 | 53.65 ±1.17 | 69.82 ±0.99 | **59.17** ±1.68 | **74.34** ±1.32 | 58.40 ±2.14 | 73.73 ±1.72 |
| Rural residential | 26.80 ±2.62 | 42.24 ±3.32 | 11.79 ±8.38 | 20.44 ±13.08 | 23.40 ±1.81 | 37.90 ±2.38 | 16.38 ±0.64 | 28.14 ±0.94 | 24.21 ±0.70 | 38.98 ±0.90 | **27.08** ±1.04 | **42.61** ±1.29 |
| Traffic land | 30.34 ±8.67 | 46.09 ±10.57 | 22.75 ±2.79 | 37.01 ±3.71 | 31.35 ±1.57 | 47.72 ±1.83 | 15.91 ±2.87 | 27.39 ±4.33 | 33.95 ±2.62 | 50.65 ±2.94 | **38.18** ±2.62 | **55.22** ±2.71 |
| Irrigated land | 0.58 ±0.83 | 1.14 ±1.63 | 2.57 ±1.22 | 4.96 ±2.31 | 3.25 ±2.74 | 6.22 ±5.08 | 2.95 ±2.26 | 5.67 ±4.21 | **6.03** ±4.27 | **11.18** ±7.43 | 1.35 ±1.11 | 2.65 ±2.15 |
| Garden plot | 0 | 0 | 0 | 0 | 1.53 ±1.35 | 3.00 ±2.64 | **2.65** ±3.44 | **5.01** ±6.43 | 0.04 ±0.07 | 0.04 ±0.14 | 0.04 ±0.05 | 0.08 ±0.10 |
| Arbor woodland | **61.01** ±1.04 | **75.78** ±0,81 | 53.72 ±7.85 | 69.66 ±6.76 | 58.00 ±3.91 | 73.37 ±3.17 | 49.66 ±10.76 | 65.89 ±9.81 | 59.32 ±1.98 | 74.46 ±1.57 | 56.73 ±5.12 | 73.92 ±6.87 |
| Shrub land | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Lake | 27.79 ±4.41 | 43.37 ±5.47 | 25.49 ±4.47 | 40.48 ±5.99 | **30.71** ±9.49 | **45.79** ±11.69 | 28.33 ±4.11 | 44.05 ±4.99 | 26.28 ±13.77 | 40.28 ±18.54 | 28.20 ±7.09 | 44.70 ±7.14 |
| Pond | **3.55** ±5.10 | **6.55** ±9.92 | 0.06 ±0.10 | 0.11 ±0.20 | 3.21 ±3.99 | 6.03 ±7.35 | 0.85 ±1.23 | 1.67 ±2.40 | 2.57 ±2.46 | 4.94 ±4.69 | 1.58 ±2.73 | 3.02 ±5.22 |
| Others | 64.95 ±0.08 | 78.76 ±0.06 | 65.13 ±2.45 | 78.86 ±1.80 | 63.66 ±1.18 | 77.79 ±0.89 | 64.50 ±1.27 | 78.41 ±0.94 | 66.57 ±0.83 | 79.93 ±0.60 | **66.65** ±1.00 | **78.40** ±3.37 |
| OA | 72.61±0.53 | | 68.41±3.41 | | 71.34±1.20 | | 69.49±1.64 | | **73.58**±0.32 | | 73.18±0.76 | |
| mIOU | 32.38±0.44 | | 27.29±1.72 | | 31.95±1.30 | | 32.89±0.84 | | **32.89**±0.84 | | 32.51±0.12 | |
| Macro-F1 | 38.84±0.78 | | 32.84±1.91 | | 38.94±1.39 | | 35.02±1.57 | | **39.53**±0.96 | | 39.10±0.34 | |

scattered distribution of mining area features, GCNs can improve the expression ability of these features. GCNs rely on an adjacency matrix for feature propagation, so the construction method of the adjacency matrix is important. Different from the common construction method of adjacency matrix (constructing edges between adjacent nodes and using gradient operators to construct edges), in a long-range information module, the input features are projected to a new dimension, the convolution neural network is used to learn the construction of adjacency matrix, and the feature propagation is completed in this dimension and then reprojected to the original dimension. The results also showed the effectiveness of the long-range information module in LCMA.

### B. *Effects of EG-UNet and Comparison Models*

In our proposed dataset, we briefly discussed the characteristics of the comparison models in Sections I-B and IV. As noted in Section V-A, models with excessive layers,

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TABLE IX
QUANTITATIVE COMPARISON OF IMAGE B IN TEST DATASET OF GID

| Class | UNet | | Attention UNet | | Deeplab v3+ | | SwinUNet | | DE-DCGCN-EE | | EG-UNet | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | IOU | F1 | IOU | F1 | IOU | F1 | IOU | F1 | IOU | F1 | IOU | F1 |
| Industrial land | 31.55 ±5.09 | 47.81 ±6.03 | 21.97 ±1.69 | 36.00 ±2.25 | 36.35 ±4.30 | 53.22 ±4.60 | 33.78 ±4.07 | 50.41 ±4.63 | **40.43** ±5.71 | **57.42** ±5.78 | 33.12 ±1.72 | 49.74 ±1.92 |
| Urban residential | 41.41 ±5.42 | 58.43 ±5.48 | 39.19 ±3.81 | 56.24 ±3.92 | 39.86 ±3.16 | 56.95 ±3.26 | 30.23 ±3.56 | 46.34 ±4.14 | 44.47 ±3.91 | 61.49 ±3.80 | **44.71** ±4.46 | **61.69** ±4.31 |
| Rural residential | 12.56 ±6.62 | 21.89 ±10.84 | 12.14 ±6.08 | 21.31 ±9.41 | 16.84 ±5.40 | 28.59 ±7.78 | 14.25 ±1.25 | 24.93 ±1.92 | **22.23** ±4.42 | **36.23** ±5.80 | 16.77 ±6.30 | 28.35 ±9.10 |
| Traffic land | 28.51 ±7.36 | 44.03 ±9.20 | 19.27 ±1.58 | 32.29 ±2.20 | 28.08 ±2.04 | 43.82 ±2.47 | 18.07 ±1.14 | 30.60 ±1.65 | 32.50 ±1.37 | 49.05 ±1.56 | **32.69** ±1.65 | **49.25** ±1.86 |
| Irrigated land | 23.19 ±10.16 | 36.89 ±13.93 | 34.89 ±3.17 | 51.67 ±3.52 | 24.30 ±7.37 | 38.72 ±9.51 | 23.64 ±6.49 | 37.94 ±8.46 | **42.86** ±9.00 | **59.64** ±8.72 | 32.00 ±4.79 | 48.33 ±5.51 |
| Dry cropland | 0.15 ±0.26 | 0.30 ±0.52 | 0.46 ±0.64 | 0.91 ±1.27 | 0.85 ±1.47 | 1.71 ±2.81 | **3.78** ±4.30 | **7.06** ±7.83 | 0.02 ±0.03 | 0.03 ±0.06 | 0.45 ±0.85 | 0.88 ±1.68 |
| Arbor woodland | **14.21** ±7.74 | **24.37** ±11.48 | 4.78 ±4.94 | 8.84 ±8.93 | 10.88 ±5.32 | 19.35 ±8.72 | 2.65 ±2.07 | 5.10 ±3.89 | 3.76 ±2.49 | 7.17 ±4.69 | 10.65 ±8.22 | 18.55 ±12.72 |
| Artificial meadow | 0.07 ±0.12 | 0.13 ±0.23 | 0 | 0 | **0.89** ±1.54 | **1.73** ±3.00 | 0 | 0 | 0 | 0 | 0.24 ±0.44 | 0.48 ±0.88 |
| River | 29.04 ±9.52 | 44.42 ±11.89 | 29.97 ±10.75 | 45.17 ±13.41 | 41.23 ±4.53 | 58.28 ±4.63 | 31.65 ±12.98 | 47.11 ±14.87 | 20.23 ±12.14 | 32.49 ±17.26 | **56.67** ±12.99 | **70.84** ±11.03 |
| Others | 42.86 ±2.19 | 59.98 ±2.13 | 44.62 ±0.81 | 61.70 ±0.77 | 42.67 ±0.60 | 59.82 ±0.60 | 40.37 ±1.09 | 57.51 ±1.11 | **44.65** ±0.53 | **61.73** ±0.50 | 43.20 ±0.55 | 60.34 ±0.54 |
| OA | 51.80±3.72 | | 51.52±1.57 | | 52.56±0.35 | | 47.35±0.65 | | 55.32±1.20 | | **56.43**±0.56 | |
| mIOU | 25.03±3.99 | | 23.51±0.99 | | 26.77±1.14 | | 22.34±1.31 | | 27.86±1.51 | | **29.63**±0.05 | |
| Macro-F1 | 33.83±5.21 | | 31.43±1.03 | | 36.22±1.71 | | 30.70±1.76 | | 36.52±1.96 | | **38.84**±0.84 | |

particularly pooling layers, are unsuitable for the proposed dataset. As PSPNet (backbone: ResNet-101) and DeepLab v3+ (backbone: ResNet-101) use deep CNNs as backbones, they exhibited poor performance (the details are presented in Table V). Attention U-Net is mainly applied to the semantic segmentation of medical images, and the attention mechanism is used to weaken the background information. However, this may result in a loss of class information in multilabel classification (the IOU of vegetable and fruit greenhouse, shrub forest, mine pit pond, red roof, and blue roof was 0%). For the few sample classes, there is no designed module in Swin UNet, which leads to its poor performance. It is worth noting that HRNet and ResUNet exhibited good performance. The results demonstrate that the dense connections of the multifeature fusion and the residual structure are effective for the pixelwise semantic segmentation task.

In GID, we verified the generalization ability of the proposed model. According to Tables VIII and IX, DE-DCGCN-EE and the proposed method performed better, especially for classes with few samples. Although DE-DCGCN-EE did not perform well in our dataset, it showed the best OA and mIOU of image A in the test dataset. Therefore, when there are enough samples, the edge enhancement strategy is helpful to improve the model accuracy for classes with few samples.

It is noteworthy that DE-DCGCN-EE also applies edge enhancement and GCNs to vessel segmentation in medical images. The network uses eight basic residual structures as backbones. Unlike our EE module, it uses four kinds of Sobel operators for edge extraction and adds these edge features to the original features. Furthermore, different from our long-range information module, when applying GCNs, DE-DCGCN-EE performs the average pooling operation on the input features and resizes the output features as the adjacency matrix.

Our model performed better in our proposed LCMA dataset, but, in GID, DE-DCGCN-EE performs better in image A. We think it is mainly because of the different usage of edge information. Our model uses edge information as a weight matrix, which can strengthen the features of irregular and small objects, and weaken the features of regular and big objects. In contrast, DE-DCGCN-EE adds edge information to the original features, which only strengthens the features of irregular and small objects. Roads (or traffic land) and mining area targets have similar characteristics. Our model shows better results in road segmentation on GID and our LCMA dataset, which can verify that our edge enhancement strategy is more effective when classifying irregular and small objects. However, because DE-DCGCN-EE does not weaken the features of regular and big objects, its overall performance is better than our proposed model in GID.

### C. Limitations and Scope for Improvement

The proposed model exhibits the following limitations.

*1) Limitations of the GCN Module in EG-UNet:* In the early stages of designing the GCN, we believed that it was preferable to apply the GCN to propagate the original input features, but, when considering each pixel as a node in the graph, the size of the adjacency matrix was 65 536 × 65 536 pixels (corresponding to an input image size of 256 × 256 pixels). Such large values require considerable computational power, far beyond the available computing power. Hence, we placed the GCN module after the final downsampling stage. Although the results indicate that the GCN can increase the accuracy, we suggest that applying the GCN to propagate the original features will yield the best results, which will be verified in future studies.

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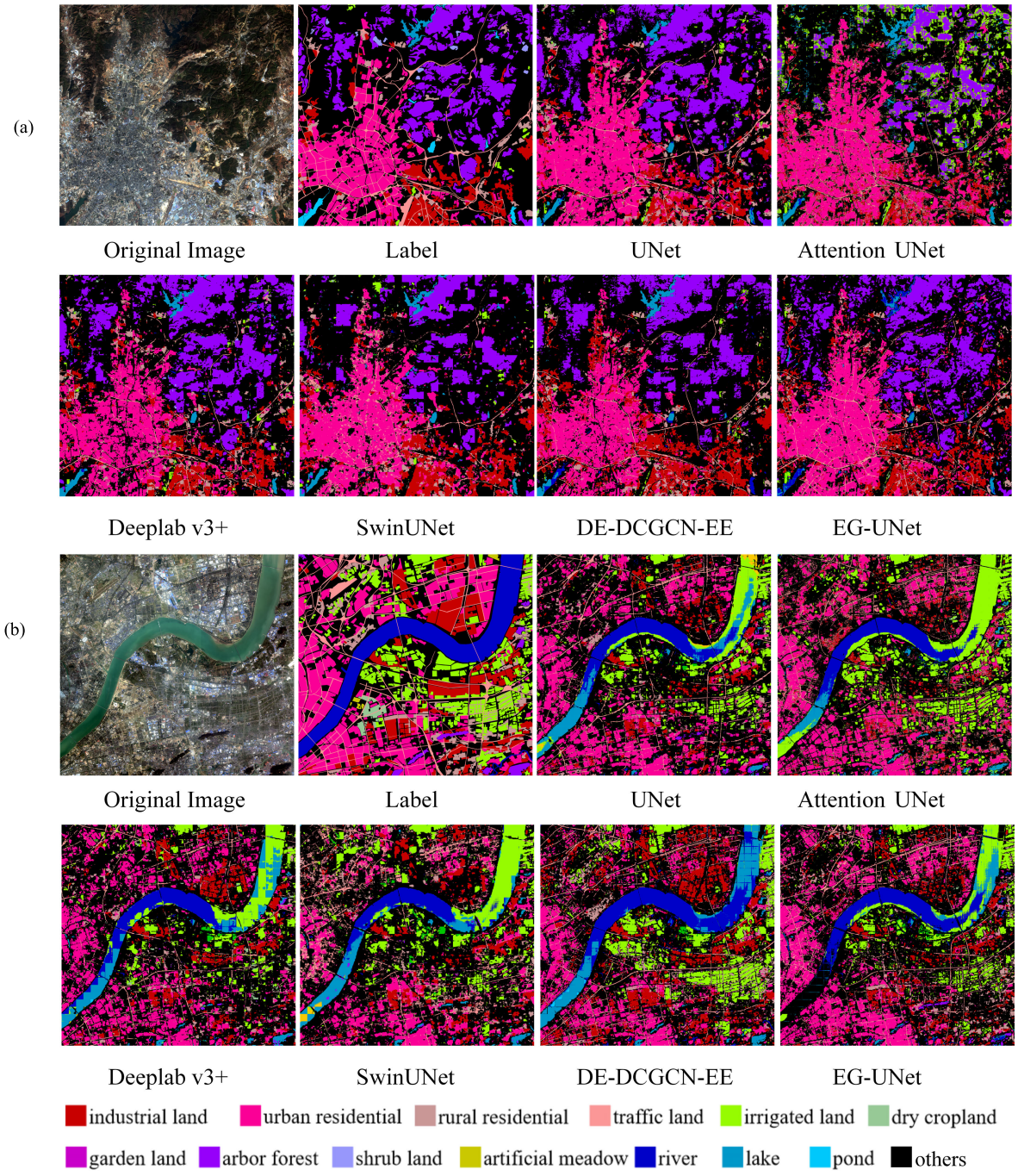


Fig. 12. Visual comparison of different models about image (a) and (b) in the test dataset of GID [73].

*2) Limitations of the Proposed Dataset:* As the generation of a dataset is time-consuming and laborious, we used only a small fraction of the remote sensing images in the dataset, which may limit the regional generalization ability of the model. Moreover, the sample numbers of certain classes were minimal, which affected the sample distribution significantly.

We will implement several targeted improvements to address the above limitations in future work.

1) *Model Improvement:* The dense connections of the multifeature fusion and residual structure may improve the classification accuracy in the study area. We will consider this aspect in the subsequent model improvement. Furthermore, inspired by DE-DCGCN-EE, we will integrate different edge extraction methods in future work.
2) *Dataset Improvement:* Considering the limitations of the proposed dataset, we will expand the research area, enhance the samples in the dataset, and release version 2.0 of the dataset.

## VI. Conclusion

Fine-scaled LCMA is important in environment assessment. Hence, we used a fine-scale land cover semantic segmentation

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dataset in this study to provide more information on mining areas. We constructed the EG-UNet model to segment objects with sparse spatial distributions and irregular shapes in mining areas.

We proposed an EE module to enhance the boundary features to avoid the problem of feature loss in extracting features of mining areas. To solve the problem of weak representation of the extracted features, we constructed a GCN module to capture long-range information between objects of the same type. U-Net was used as the backbone of the proposed model. When compared with several classical and recent models, the following conclusions can be drawn: 1) the EE and GCN modules help classify classes with few samples and 2) the proposed model performs better than other models, particularly for classes with few samples. The innovations of this study are given as follows: 1) we designed an EE module to enhance features and 2) we constructed a GCN module to capture long-range information.

We plan to evaluate the effects of the dense connections of multifeature fusion and the residual structure to enhance the ability to classify each category further. Moreover, we aim to improve our dataset in future studies.

## VII. Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

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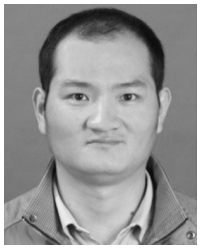

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